

Set: A set is a collection of objects.

The objects in a set are called elements of the set.

A well-defined set is a set in which we know for sure if an element belongs to that set.

Example:

— The set of all movies in which Aamir Khan appears is well-defined.

— The set of all movie serials made by Republic Pictures is well-defined.

— The set of best TV shows of all time is not well-defined.

Notation:

Capital letters $A, B, C, \dots$ are ordinarily used to denote sets and lower case letters $a, b, c, \dots$ to denote elements of sets.

Representation of a set:

There are two ways of representation a set.

1. Roster or Tabular form
2. Rule Method or Set builder form

Roster or Tabular form

In this form all the elements of the set are listed, the elements being separated by commas and are enclosed within braces.

Example:

Let C = The set of all movies in which Aamir Khan appears.
 $C = \{PK, 3 Idiots, Lagaan, Tare Zameen Par\}$

Example: Let set $A =$ The set of odd numbers greater than zero, and less than 10.

$$A = \{1, 3, 5, 7, 9\}.$$

Rule Method or Set builder form:

When it is not convenient to list all the elements of a set, we use a notation that employs the rules in which an element is a member of the set. This is called set-builder form.

Example: 1, $V = \{ \text{people} \mid \text{citizens registered to vote in India} \}$

2, $A = \{ x \mid x > 5 \} =$ This is the set A that has all real numbers greater than 5.

Special Sets of Numbers:

$N =$ The set of natural numbers.

$$= \{1, 2, 3, \dots\}.$$

$W =$ The set of whole numbers.

$$= \{0, 1, 2, 3, \dots\}$$

$Z =$ The set of integers

$$= \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

$Q =$ The set of rational numbers.

$$= \{x \mid x = p/q, \text{ where } p \text{ and } q \text{ are elements of } Z \text{ and } q \neq 0\}$$

$H =$ The set of irrational numbers.

$R =$ The set of real numbers.

$C =$ The set of complex numbers.

Finite Set:

A set which contains a definite number of elements is called a finite set.

- Example: 1, The set of all colors in the rainbow.
2, The set of students in a class.
3, $P = \{2, 3, 5, 7, 11, 13, 17, \dots, 97\}$.

Infinite Set:

The set whose elements cannot be listed, i.e., set containing never-ending elements is called an infinite set.

- Example: 1, $A = \text{set of integers} = \{0, 1, 2, \dots\}$
2, $B = \{1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \dots\}$.

Empty Set:

A set which does not contain any elements is called as Empty set or Null set or Void set. It is denoted by ϕ or $\{\}$.

- Example: 1, The set of whole numbers less than 0.
2, $A = \{x \mid x \in \mathbb{N}, 3 < x < 4\}$.

Singleton Set:

A set which has only one element is called a Singleton set.

- Example: 1, let $B = \{x : x \text{ is an even prime number}\}$
2, $A = \{x : x \text{ is neither prime nor composite}\}$

Sub Set: If every element of a set A is also an element of set B , we say set A is a subset of set B .
 $A \subseteq B$, if $x \in A$ and $x \in B$.

Example:
1) If $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{1, 2, 3, 4\}$
Then $B \subseteq A$.

2) If $A = \{6, 5, 4\}$ and $B = \{4, 5, 6\}$ then
 $A \subseteq B$ and $B \subseteq A$.

Proper Subset: Any subset A is said to be proper subset of another set B if A is a subset of B , but there is at least one element of B which does not belong to A i.e.,
if $A \subseteq B$ but $A \neq B$. It is written as $A \subset B$.

Example: $A = \{1, 5\}$, $B = \{1, 5, 6\}$, $C = \{1, 6, 5\}$
Here, $A \subseteq C$, $B \subseteq C$, and $A \subset C$, whereas
 $B \not\subset C$ since $B = C$.

Equal Set: Two sets A and B are called equal if they have equal numbers and similar types of elements.
i.e., $A \subseteq B$ and $B \subseteq A \Rightarrow A = B$

Example: If $A = \{1, 3, 4, 5, 6\}$, $B = \{4, 1, 5, 6, 3\}$
Here $A \subseteq B$ and $B \subseteq A$ So $A = B$.

Universal Set: A set which contains all the elements of other given sets is called a Universal set. The symbol for denoting a universal set is U .

Example: $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, $C = \{3, 5, 7\}$

- Then $U = \{1, 2, 3, 4, 5, 7\}$.

Number of Subsets of a Set:

If a set contains 'n' elements, then the number of subsets of the set is 2^n .

Example: List all the subsets of the set $A = \{a, b, c\}$.

The given set contains 3 elements, hence it has $2^3 = 8$ subsets.

The subsets are ϕ , $\{a\}$, $\{b\}$, $\{c\}$, $\{a, b\}$, $\{a, c\}$, $\{b, c\}$, $\{a, b, c\}$.

Number of Proper Subsets of the set:

If a set contains 'n' elements, then the number of proper subsets of the set is $2^n - 1$.

Example: $A = \{p, q\}$

Number of proper subsets of A are $2^2 - 1 = 4 - 1 = 3$.

$\{\}, \{p\}, \{q\}$.

Cardinal Number of a Set:

The number of distinct elements in a given set A is called the cardinal number of A. It is denoted by $n(A)$.

Example: 1, $A = \{x : x \in \mathbb{N}, x < 5\}$

$A = \{1, 2, 3, 4\}$ Therefore, $n(A) = 4$.

2, $B =$ set of letters in the word ALGEBRA

$B = \{A, L, G, E, B, R\}$ Therefore, $n(B) = 6$.

Power Set: The collection of all subsets of a set A is called the power set of A . It is denoted by $P(A)$.

In $P(A)$, every element is a set.

Example: If $A = \{p, q\}$ then all the subsets of A will be

$\phi, \{p\}, \{q\}, \{p, q\}$

$$P(A) = \{ \phi, \{p\}, \{q\}, \{p, q\} \}$$

Number of elements of $P(A) = n[P(A)] = 2^m$
where m = no. of elements in set A .

Lecture # 2

Operations on Sets

- Union
- Intersection
- Complements
- Difference
- Symmetric Difference.

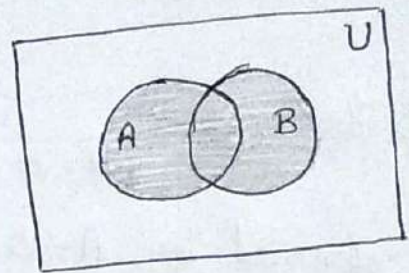
Union: let A and B are two sets. The union of the sets A and B , denoted by $A \cup B$, is the set:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

Example: $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$

$$A \cup B = \{1, 2, 3, 4, 5\}$$

Venn Diagram for $A \cup B$.



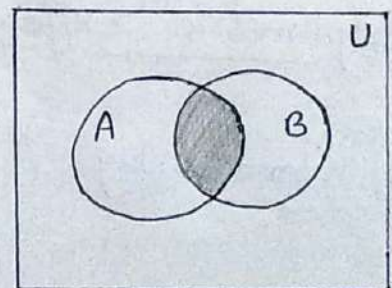
Intersection: The intersection of two sets A and B , denoted by $A \cap B$, is

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

Example: $A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$

$$A \cap B = \{3\}$$

Venn Diagram for $A \cap B$.



Theorem: for any sets A and B ,

(i) $A \cap B \subseteq A \subseteq A \cup B$ and (ii) $A \cap B \subseteq B \subseteq A \cup B$.



Complement: If A is a set, then the complement of the A (with respect to U), denoted by $\bar{A}$ is the set $U - A$

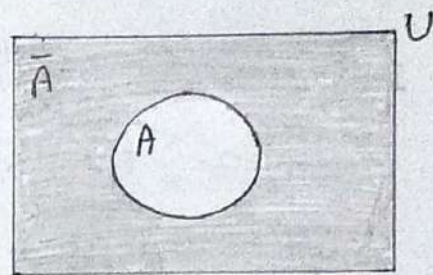


$$A^c = \{x \mid x \in U, x \notin A\}$$

Example: If U is the positive integers less than 100, what is the complement of $\{x \mid x > 70\}$

$$\bar{A} = \{x \mid x \leq 70\}$$

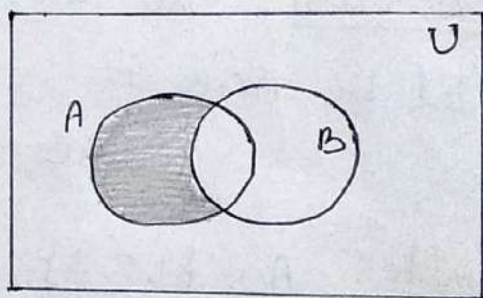
Venn Diagram for Complement.



Difference: Let A and B be sets. The difference of A and B , denoted by $A - B$, is the set containing the elements of A that are not in B .

$$A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

Venn Diagram for $A - B$.



Symmetric Difference: The symmetric difference of A and B , denoted by $A \Delta B$ is the set

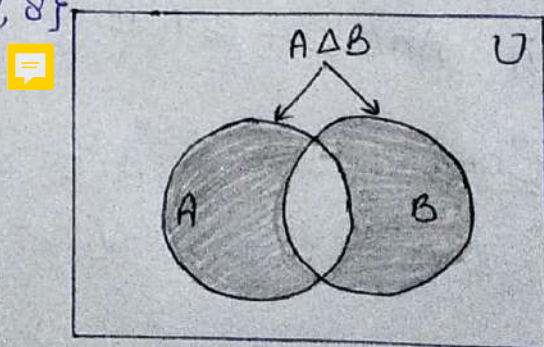
$$A \Delta B = (A - B) \cup (B - A)$$

Example. $U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

$$A = \{1, 2, 3, 4, 5\}, B = \{4, 5, 6, 7, 8\}$$

$$A \Delta B = \{1, 2, 3, 6, 7, 8\}$$

Venn Diagram for $A \Delta B$



Laws of the algebra of Sets.

Idempotent Laws:

$$1a) A \cup A = A$$

$$1b) A \cap A = A$$

Associative Laws:

$$2a) (A \cup B) \cup C = A \cup (B \cup C)$$

$$2b) (A \cap B) \cap C = A \cap (B \cap C)$$

Commutative Laws:

$$3a) A \cup B = B \cup A$$

$$3b) A \cap B = B \cap A$$

Distributive Laws:

$$4a) A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$4b) A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Identity Laws:

$$5a) A \cup \phi = A$$

$$5b) A \cap U = A$$

$$6a) A \cup U = U$$

$$6b) A \cap \phi = \phi$$

Involution Law:

$$7) (A^c)^c = A$$

Complement Laws:

$$8a) A \cup A^c = U$$

$$8b) A \cap A^c = \phi$$

$$9a) U^c = \phi$$

$$9b) \phi^c = U$$

De Morgan's Laws:

$$10a) (A \cup B)^c = A^c \cap B^c$$

$$10b) (A \cap B)^c = A^c \cup B^c$$

Proof of Distributive Law:

$$4a, \quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$\begin{aligned} \text{Let } x \in A \cup (B \cap C) &\Rightarrow x \in A \text{ or } x \in (B \cap C) \\ &\Rightarrow x \in A \text{ or } (x \in B \text{ and } x \in C) \\ &\Rightarrow (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C) \\ &\Rightarrow x \in (A \cup B) \text{ and } x \in (A \cup C) \\ &\Rightarrow x \in (A \cup B) \cap (A \cup C) \end{aligned}$$

$$\text{Thus, } A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C) \quad \text{--- (i)}$$

Conversely, Let

$$\begin{aligned} x \in (A \cup B) \cap (A \cup C) &\Rightarrow x \in (A \cup B) \text{ and } x \in (A \cup C) \\ &\Rightarrow (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C) \\ &\Rightarrow x \in A \text{ or } (x \in B \text{ and } x \in C) \\ &\Rightarrow x \in A \cup (B \cap C) \end{aligned}$$

$$(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C) \quad \text{--- (ii)}$$

$$\text{From (i) and (ii) } A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \quad \underline{\text{Proved}}$$

Proof of DeMorgan's Law:

$$10a, \quad (A \cup B)' = A' \cap B'$$

$$\begin{aligned} \text{Let } x \in (A \cup B)' &\Rightarrow x \notin (A \cup B) \\ &\Rightarrow x \notin A \text{ and } x \notin B \\ &\Rightarrow x \in A' \text{ and } x \in B' \\ &\Rightarrow x \in (A' \cap B') \end{aligned}$$

$$\text{Hence } (A \cup B)' \subseteq (A' \cap B') \quad \text{--- (i)}$$

Conversely, Let

$$\begin{aligned} x \in A' \cap B' &\Rightarrow x \in A' \text{ and } x \in B' \\ &\Rightarrow x \notin A \text{ and } x \notin B \end{aligned}$$

$$\Rightarrow x \notin (A \cup B)$$

$$\Rightarrow x \in (A \cup B)'$$

Thus, $A' \cap B' \subseteq (A \cup B)'$ — (ii)

From (i) and (ii) $(A \cup B)' = \underline{A' \cap B'}$ Proved.

Properties of Relations:

Reflexive Relation: A relation R on a set A is reflexive if aRa for every $a \in A$, that is, if $(a, a) \in R$ for every $a \in A$. This simply means that each element a of A is related to itself.

For example,

(a) If $R_1 = \{(1,1), (1,2), (2,2), (2,3), (3,3)\}$ be a relation on $A = \{1,2,3\}$, then R_1 is reflexive relation since for every $a \in A$, $(a,a) \in R_1$.

(b) If $R_2 = \{(1,1), (1,2), (2,3), (3,3)\}$ be a relation on $A = \{1,2,3\}$, then R_2 is not a reflexive relation since for $2 \in A$, $(2,2) \notin R_2$.

(c) $R_3 = \{(x,y) \in \mathbb{R}^2 : x \leq y\}$ is a reflexive relation since $x \leq x$ for any $x \in \mathbb{R}$ (a set of real numbers).

Irreflexive Relation: A relation R on a set A is irreflexive if, for every $a \in A$, $(a,a) \notin R$. In other words, there is no $a \in A$ such that aRa . The terms reflexive and irreflexive are extreme cases. Reflexive means that aRa is true for all a , and irreflexive means that aRa is true for no a .

For example,

(a) The relation $R_1 = \{(1,2), (1,3), (2,1), (2,3)\}$ on $A = \{1,2,3\}$ is irreflexive relation since $(x,x) \notin R_1$ for every $x \in R_1$.

(b) The relation $R_2 = \{(x,y) \in \mathbb{R}^2 : x < y\}$ is an irreflexive relation since $x < x$ for no $x \in \mathbb{R}$.

Non-reflexive Relation: A relation R on a set A is non-reflexive if R is neither reflexive nor irreflexive i.e., if aRa is true for some a and false for others.

For example,

$R = \{(1,2), (2,2), (2,3), (3,1)\}$ on $A = \{1,2,3\}$ is a non-reflexive relation since $2R2$ is true but $1R1$ and $3R3$ are false.

Symmetric Relation: A relation R on a set A is symmetric if whenever $(a,b) \in R$ then $(b,a) \in R$, i.e., if $aRb \Rightarrow bRa$. This means if any one element is related to any other element, then the second element is related to the first.

For example,

(a) $R_1 = \{(1,1), (1,2), (1,3), (2,2), (2,1), (3,1)\}$ on $A = \{1,2,3\}$ is a symmetric relation.

(b) $R_2 = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ is a symmetric relation on $\mathbb{R}^2$ since if $x^2 + y^2 = 1$ then $y^2 + x^2 = 1$ too, i.e.,

if $(x,y) \in R_2$ then $(y,x) \in R_2$.

Asymmetric Relation: A relation R on a set A is asymmetric if whenever $(a,b) \in R$ then $(b,a) \notin R$ for $a \neq b$ i.e., if $aRb \Rightarrow b \not R a$. This means that the presence of (a,b) in R excludes the possibility of presence of (b,a) in R .

For example,

The relation $R_1 = \{(1,1), (2,1), (2,3), (3,1)\}$ on $A = \{1,2,3\}$ is an asymmetric relation.

Antisymmetric Relation: A relation R on a set A is antisymmetric if aRb and $bRa \Rightarrow a=b$ for all $a, b \in A$.

For example,

(a) $R_1 = \{(1,2), (2,2), (2,3)\}$ on $A = \{1,2,3\}$ is an antisymmetric relation.

(b) $R_2 = \{(x,y) \in \mathbb{R}^2 \mid x \leq y\}$ is an antisymmetric relation on $\mathbb{R}$ since $x \leq y$ and $y \leq x$ implies $x=y$, then $(x,y) \in R$ and $(y,x) \in R$ implies $x=y$.

(c) $R = \{(x,y) \in \mathbb{N} \mid x \text{ is a divisor of } y\}$ is an antisymmetric relation since x divides y and y divides x implies $x=y$.

Note:

Antisymmetric is not the same as not symmetric. A relation may be symmetric as well as antisymmetric at the same time.

For example,

The relation $R = \{(1,1), (3,3)\}$ is both symmetric and antisymmetric on $A = \{1,2,3\}$.

Transitive Relation: A relation R on a set A is transitive if whenever $(a,b) \in R$ and $(b,c) \in R$, then $(a,c) \in R$, i.e.,
 aRb and $bRc \Rightarrow aRc$.

For example:

(a) The relation 'is parallel to' on the set of lines in a plane is transitive, because if a line x is parallel to the line y and if y is parallel to line z , then x is parallel to z .

(b) The relation 'is less than' and 'is greater than' are transitive relations on the set of real numbers. If $a < b$ and $b < c$ implies $a < c$ and if $a > b$ and $b > c \Rightarrow a > c$ $\forall$ real numbers a, b, c .

The following table summarizes the above properties.

Property	Meaning
1. Reflexivity	$(a, a) \in R$, i.e. $aRa \forall a \in A$
2. Irreflexivity	$(a, a) \notin R$, i.e. $a \not R a \forall a \in A$
3. Symmetry	$(a, b) \in R \Rightarrow (b, a) \in R$, i.e. $aRb \Rightarrow bRa \forall a, b \in A$
4. Asymmetry	$(a, b) \in R \Rightarrow (b, a) \notin R$, i.e. $aRb \Rightarrow b \not R a \forall a, b \in A$
5. Antisymmetry	$(a, b) \in R \wedge (b, a) \in R \Rightarrow a = b$, i.e. aRb and $bRa \Rightarrow a = b \forall a, b \in A$
6. Transitivity	$(a, b) \in R \wedge (b, c) \in R \Rightarrow (a, c) \in R$, i.e. aRb and $bRc \Rightarrow aRc \forall a, b, c \in A$

Equivalence Relation :

A relation on a set A is called an equivalence relation or RST relation if it is reflexive, symmetric and transitive. i.e., R is an equivalence relation on A if it has the following three properties:

1. $(a, a) \in R \forall a \in A$ (reflexive)
2. $(a, b) \in R$ implies $(b, a) \in R$ (symmetric)
3. (a, b) and $(b, c) \in R \Rightarrow (a, c) \in R$ (transitive)

Example 1. If R be a relation in the set of integers $\mathbb{Z}$ defined by $R = \{ (x, y) : x \in \mathbb{Z}, y \in \mathbb{Z}, (x-y) \text{ is divisible by } 6 \}$

Then prove that R is an equivalence relation.

Sol. Let $x \in \mathbb{Z}$. Then $x - x = 0$ and 0 is divisible by 6.

Therefore, $xRx \forall x \in \mathbb{Z}$.

Hence, R is reflexive.

$$\begin{aligned}
 \text{Again, } xRy &\Rightarrow (x-y) \text{ is divisible by } 6 \\
 &\Rightarrow -(x-y) \text{ is divisible by } 6 \\
 &\Rightarrow (y-x) \text{ is divisible by } 6 \\
 &\Rightarrow yRx.
 \end{aligned}$$

Hence, R is symmetric.

$$\begin{aligned}
 xRy \text{ and } yRz &\Rightarrow (x-y) \text{ is divisible by } 6 \text{ and } (y-z) \text{ is divisible by } 6 \\
 &\Rightarrow [(x-y) + (y-z)] \text{ is divisible by } 6 \\
 &\Rightarrow (x-z) \text{ is divisible by } 6 \\
 &\Rightarrow xRz.
 \end{aligned}$$

Hence R is transitive.

Thus R is an equivalence relation.

Example 2. For any two real numbers θ and ϕ , we define $\theta R \phi$ if and only if $\sec^2 \theta - \tan^2 \phi = 1$. Show that R is an equivalence relation.

Sol. Reflexive

$\theta R \theta$ if and only if $\sec^2 \theta - \tan^2 \theta = 1$ which is true for all $\theta \in \mathbb{R}$ hence R is reflexive.

Symmetric

$$\begin{aligned}
 \theta R \phi &\Rightarrow \sec^2 \theta - \tan^2 \phi = 1 \\
 &\Rightarrow 1 + \tan^2 \theta - (\sec^2 \phi - 1) = 1 \\
 &\Rightarrow 1 + \tan^2 \theta - \sec^2 \phi + 1 = 1 \\
 &\Rightarrow \sec^2 \phi - \tan^2 \theta = 1 \\
 &\Rightarrow \phi R \theta
 \end{aligned}$$

Hence R is symmetric.

Transitive

$$\begin{aligned} \theta R \phi \text{ and } \phi R \psi &\Rightarrow \sec^2 \theta - \tan^2 \phi = 1 \text{ and } \sec^2 \phi - \tan^2 \psi = 1 \\ &\Rightarrow \sec^2 \theta - \tan^2 \phi + \sec^2 \phi - \tan^2 \psi = 2 \\ &\Rightarrow \sec^2 \theta + (\sec^2 \phi - \tan^2 \phi) - \tan^2 \psi = 2 \\ &\Rightarrow \sec^2 \theta + 1 - \tan^2 \psi = 2 \\ &\Rightarrow \sec^2 \theta - \tan^2 \psi = 1 \\ &\Rightarrow \theta R \psi \end{aligned}$$

Hence R is transitive.

Thus R is an equivalence relation.

Partial Order Relation:

A relation R on a set S is called a partial order if it is reflexive, antisymmetric and transitive. That is.

1. Reflexive: $aRa \forall a \in S$
2. Antisymmetric: $aRb \text{ and } bRa \Rightarrow a=b$
3. Transitive: $aRb \text{ and } bRc \Rightarrow aRc$.

A set S together with a partial order R is called a partial order set or a poset and is denoted by (S, R) .

For example,

The greater or equal ($\geq$) relation is a partial ordering on $\mathbb{Z}$, the set of integers.

Reflexive: Since $a \geq a$ for every integer a , $\geq$ is reflexive.

Antisymmetric: Since $a \geq b$ & $b \geq a \Rightarrow a=b$, $\geq$ is antisymmetric.

Transitive: Since $a \geq b$ & $b \geq c \Rightarrow a \geq c$, $\geq$ is transitive.

Hence, $\geq$ is a partial ordering on $\mathbb{Z}$, and $(\mathbb{Z}, \geq)$ is a poset.

Sol Let $A = \{1, 2, 3, 4, 5, 6\}$

Then $R = \{(i, j) : |i - j| = 2\}$ on A

$$= \{(1, 3), (2, 4), (3, 1), (4, 2), (3, 5), (5, 3), (4, 6), (6, 4)\}$$

R is not reflexive since $(i, i) \notin R \forall i \in A$

For eg. $(1, 1) \notin R$.

R is symmetric since for all $(i, j) \in R$, $(j, i) \in R$.

R is not transitive since for all (i, j) and $(j, k) \in R$, (i, k) does not belong to R . For eg.,

$(2, 4)$ and $(4, 2) \in R$ but $(2, 2) \notin R$.

Hence R is not transitive.

Partial Order Relation

A relation R on a set S is called a partial order if it is reflexive, antisymmetric and transitive. That is

1. Reflexive: aRa for all $a \in S$
2. Antisymmetric: aRb and $bRa \Rightarrow a = b$
3. Transitive: aRb and $bRc \Rightarrow aRc$

A set S together with a partial order R is called a partial order set or a poset and is denoted by (S, R) .

For example, the greater or equal ($\geq$) relation is a partial ordering on $\mathbb{Z}$, the set of integers.

Reflexive: Since $a \geq a$ for every integer a , $\geq$ is reflexive.

Antisymmetric: Since $a \geq b$ and $b \geq a$ imply $a = b$, $\geq$ is antisymmetric.

Transitive: Since $a \geq b$ and $b \geq c$ imply $a \geq c$, $\geq$ is transitive.

Hence, $\geq$ is a partial ordering on $\mathbb{Z}$, and $(\mathbb{Z}, \geq)$ is a poset.

(20)

Ex: Let $/$ be the divides relation R on a set N of positive integers. That is, for all $a, b \in N$, $a/b \Leftrightarrow b = ka$ for some integer k . Prove that $/$ is a partial order relation on N .

Sol. Reflexive: We have, $a \in N$, a is a divisor of a , i.e., aRa . Therefore, R is reflexive.

Antisymmetric: If a is a divisor of b then b is not be a divisor of a unless $a = b$. Thus, aRb and bRa imply $a = b$. Therefore, R is antisymmetric.

Transitive: Finally, a is a divisor of b and b is a divisor of c implies a is a divisor of c . Therefore, R is transitive.

Since R is reflexive, antisymmetric and transitive, therefore, R is a partial order relation.

Observe that on the set of all integers, the above relation is not a partial order set as a and $(-a)$ both divide each other without being equal.

Representation of Relations

Graphs of Relations: Let A and B are two finite sets and R is a relation from A to B .

For graphical representation of a relation on a set, each element of the set is represented by a point. These points are called nodes or vertices.

An arc is drawn from each point to its related point. If the pair $x \in A$, $y \in B$ is in the relation, the corresponding nodes are connected by arcs called edges or arcs.

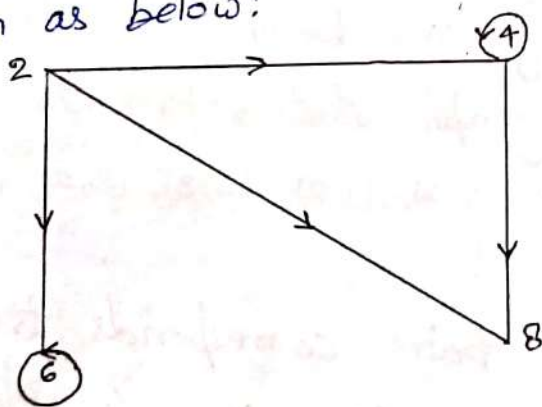
The arcs start at the first element of the pair, and they go to the second element of the pair. The direction is indicated by an arrow. All arcs with an arrow are called directed arcs. The resulting pictorial representation of R is called a directed graph or digraph of R .

An edge of the form (a, a) is represented using an arc from the vertex a back to itself. Such an edge is called loop.

For example, let $A = \{2, 4, 6\}$, $B = \{4, 6, 8\}$ and R be the relation from the set A to the set B given by:
 xRy mean x is a factor of y , then

$$R = \{(2, 4), (2, 6), (2, 8), (4, 4), (6, 6), (4, 8)\}.$$

This relation R from A to B is represented by the arrow diagram as below:



Note: that the digraph of the inverse of a relation has exactly the same edges of the digraph of the original relation, but the directions of the edges are reversed.

The directed graph representing a relation can be used to determine whether the relation has various properties.

(i) A relation is reflexive if and only if there is a loop at every vertex of the directed graph, so that ordered pair of the form (a, a) occurs in the relation.

(22)

(ii) A relation is symmetric if and only if for every edge between distinct vertices in its digraph there is an edge in the opposite direction, so that (b, a) is in the relation whenever (a, b) is in the relation.

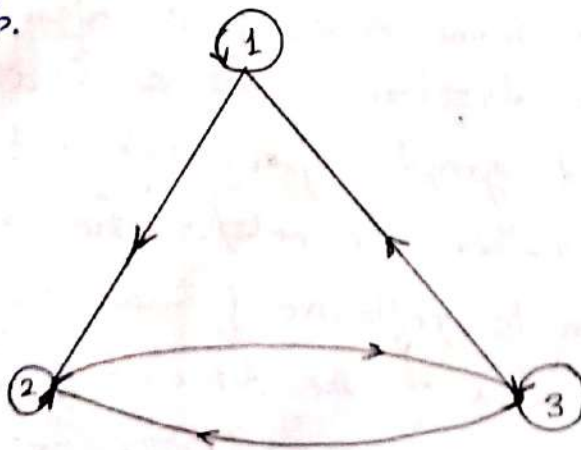
A relation is antisymmetric if no two distinct points in the digraph have an edge going between them in both directions.

(iii) A relation is transitive if and only if whenever there is a directed edge from a vertex a to a vertex b and from a vertex b to vertex c , then there is also a directed edge from a to c .

Domain and range can be found easily if the relation is represented by a graph. Every node with an outgoing arc belongs to the domain, and every node with an incoming arc belongs to the range.

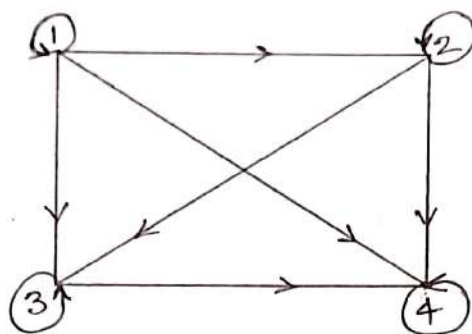
Ex: 1. Draw the digraph that represents the relation $R = \{(1, 1), (2, 2), (1, 2), (2, 3), (3, 2), (3, 1), (3, 3)\}$ on $X = \{1, 2, 3\}$.

Sol. Each of these pairs corresponds to an edge of the directed graph with $(1, 1), (2, 2), (3, 3)$ corresponding to loop.



Ex: 2 Determine whether the relation for the directed graph are reflexive, symmetric, antisymmetric and or transitive.

Sol.



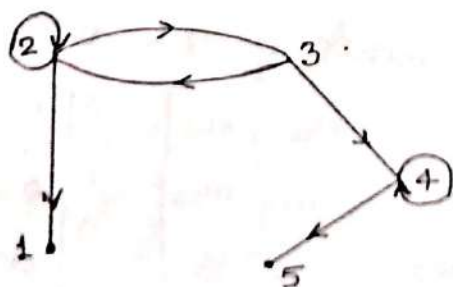
Reflexive: The relation is reflexive since the digraph of the relation has a loop at every vertex.

Symmetric: The relation is not symmetric. The digraph of the relation has a directed edge from 2 to 3, but there is no directed edge from 3 to 2. i.e., $2R3$ but $3 \not R 2$.

Antisymmetric: The relation is antisymmetric. The digraph of the relation has at most one directed edge between each pair of vertices.

Transitive: The relation is transitive. The digraph of the relation has the property whenever there are directed edges from x to y and from y to z , there is also a directed edge from x to z .

Ex: 3 Write the relation as a set of ordered pairs from the digraph:



Sol.

$$R = \{(2, 1), (2, 2), (2, 3), (3, 2), (3, 4), (4, 4), (4, 5)\}$$

The Matrix of a Relation:

Let $A = \{a_1, a_2, a_3, \dots, a_m\}$ and $B = \{b_1, b_2, b_3, \dots, b_n\}$ are finite sets containing m and n elements respectively and let R be a relation from A to B . Then R can be represented by the mn matrix

$$M_R = [m_{ij}] \text{ where}$$

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R \\ 0 & \text{if } (a_i, b_j) \notin R \end{cases}$$

The matrix M_R is called the matrix of R . The matrix M_R is a binary matrix.

Ex.1. Let R be the relation from the set $A = \{1, 3, 4\}$ on itself and defined by $R = \{(1, 1), (1, 3), (3, 3), (4, 4)\}$ then the matrix of R can be found as follows.

Let M_R denotes the matrix of R . Number of rows in $M_R = \text{No. of elements in } A = 3$. Since the relation from the set A on itself, the no. of columns in M_R is also 3. So M_R is a 3×3 matrix.

We have $a_1 = 1, a_2 = 3, a_3 = 4$

And $b_1 = 1, b_2 = 3, b_3 = 4$

Since $1R1$, we have $m_{11} = 1$ as $(a_1, b_1) = (1, 1)$

$1R3$, we have $m_{12} = 1$ as $(a_1, b_2) = (1, 3)$

$3R3$, we have $m_{22} = 1$ as $(a_2, b_2) = (3, 3)$

$4R4$, we have $m_{33} = 1$ as $(a_3, b_3) = (4, 4)$

and all other elements of M_R are zero.

Hence,
$$M_R = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Ex:2. Let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4\}$. Which ordered pairs are in the relation R represented by the matrix

$$M_R = \begin{matrix} & \begin{matrix} b_1 & b_2 & b_3 & b_4 \end{matrix} \\ \begin{matrix} a_1 \\ a_2 \\ a_3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Sol. Since R contains of those ordered pairs (a_i, b_j) with $m_{ij} = 1$. It follows that

$$R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_1), (a_3, b_3)\}$$

If the relation is given by a matrix, then the domain is given by the rows that contain at least one non-zero entry, and the range is similarly, given by the columns with at least one non-zero entry.

If R and S are relation on a set A . Using operations on Boolean matrices, one can show

$$M_{R \cup S} = M_R \vee M_S$$

$$M_{R \cap S} = M_R \wedge M_S$$

$$M_{R^{-1}} = M_R^T$$

Let $M_R = [m_{ij}]$ represents the matrix of relation R .

(ii) Reflexive: If all the elements in the main diagonal of the matrix representation of a relation are 1, then the relation is reflexive. The main diagonal elements of the matrix M_R is the set of all m_{ii} . If all $m_{ii} = 1$, then the relation is reflexive. Thus

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ is the representation of a reflexive relation } R.$$

If all $m_{ii} = 0$, then the relation is irreflexive.

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

Hence, reflexivity and irreflexivity depend only on the diagonal elements.

(ii) Symmetric: If the representative matrix of a relation is symmetric with respect to the main diagonal, i.e., $m_{ij} = m_{ji}$ for all values of i and j then the relation is symmetric (i.e., $M_R = M_R^T$).

A relation is antisymmetric if and only if $m_{ij} = 1$ necessitates that $m_{ji} = 0$.

Symmetric

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Antisymmetric

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

(iii) Transitive: There is no simple way to test whether a relation R is transitive by examining the matrix M_R . A relation R is transitive if and only if its matrix $M_R = [m_{ij}]$ has the property, if $m_{ij} = 1$ and $m_{jk} = 1$, then $m_{ik} = 1$. This statement simply means R is transitive if $M_R \cdot M_R$ has 1 in position i, k . Thus, the transitivity of R means that if $M_R^2 = M_R \cdot M_R$ has a 1 in any position then M_R must have a 1 in the same position. Thus, R is transitive if and only if $M_R^2 + M_R = M_R$.

Ex: Let $A = \{1, 2, 3, 4\}$ and let R be a relation on A whose matrix is

$$M_R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Show that R is transitive.

Sol.

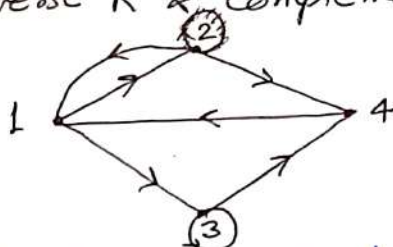
$$M_R^2 = M_R \cdot M_R$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore M_R^2 + M_R = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = M_R$$

$\therefore$ The relation R is transitive.

Ex. Consider a relation R whose directed graph is shown. Determine its inverse R^{-1} & Complement R' .

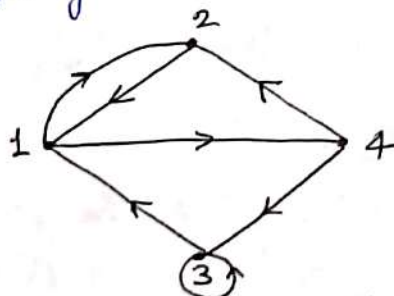


Sol. The elements of relation R from the given directed graph is $R = \{(1, 2), (1, 3), (2, 1), (3, 3), (2, 4), (3, 4), (4, 1)\}$

(i) The inverse of relation R is

$$R^{-1} = \{(2, 1), (3, 1), (1, 2), (4, 2), (3, 3), (4, 3), (1, 4)\}$$

The directed graph of R^{-1} is shown as



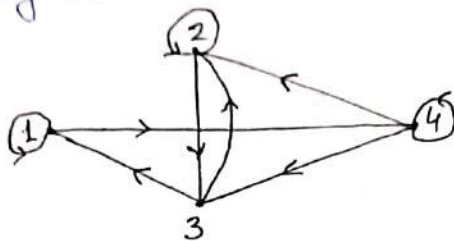
The universal relation $R \times R$ on the set $\{1, 2, 3, 4\}$ is

$$R \times R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4), (4, 1), (4, 2), (4, 3), (4, 4)\}$$

Thus, the complement of R is

$$R' = \{(1, 1), (1, 4), (2, 2), (2, 3), (3, 2), (3, 1), (4, 4), (4, 3), (4, 2)\}$$

The directed graph of R' is shown as



Ex. Consider a relation R defined on $A = \{1, 2, 3\}$ whose matrix representation is given below. Determine its inverse R^{-1} & Complement R'

$$M_R = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Sol We have

$$\begin{aligned} M_{R^{-1}} &= \text{transpose of } M_R \\ &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \end{aligned}$$

So, the inverse relation of R is

$$R^{-1} = \{(1,1), (1,2), (2,2), (3,2), (3,3)\}$$

where

$$R = \{(1,1), (2,1), (2,2), (2,3), (3,3)\}$$

Again, we know $M_{R'}$ can be obtained by changing 0 into 1 and 1 into 0 in M_R

$$\therefore M_{R'} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$R' = \{(1,2), (1,3), (3,1), (3,2)\}$$

Composition of Relations:

Let A, B, C be sets. Let R be a relation from A to B and let S be a relation from B to C . That is R is subset of $A \times B$ and S is a subset of $B \times C$. The composite of R and S , denoted by $R \circ S$, is the relation consisting of ordered pairs (a, c) when $a \in A$ and $c \in C$ and for which there exists an element $b \in B$ such that

$(a,b) \in R$ and $(b,c) \in S$. Thus,

$$R \circ S = \{ (a,c) \in A \times C : \text{for some } b \in B, (a,b) \in R \text{ and } (b,c) \in S \}$$

The composition of a relation with itself is denoted with a power of a relation R . Let R be a relation on the set A . Then $R \circ R$ is the composition of R with itself and $R \circ R$ is denoted by R^2 . Similarly, $R^3 = R^2 \circ R = R \circ R \circ R$ and so on.

Ex:1. Let $A = \{1, 2, 3\}$, $B = \{p, q, r\}$, $C = \{x, y, z\}$ and Let $R = \{(1, p), (1, r), (2, q), (3, q)\}$ and $S = \{(p, y), (q, x), (r, z)\}$. Compute $R \circ S$.

Sol. $R \circ S = \{(1, y), (1, z), (2, x), (3, x)\}$

Ex:2. Let $R = \{(1, 1), (2, 1), (3, 2)\}$, compute R^2 .

Sol. $R^2 = R \circ R = \{(1, 1), (2, 1), (3, 1)\}$

Matrix Representation of Composition:

Let A , B and C be finite sets. Let R be a relation from A to B and S be a relation from B to C . Then

$$M_{R \circ S} = M_R \cdot M_S$$

where M_R and M_S represent relation matrices of R and S respectively.

Ex: $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $M_S = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

The matrix of composite relations can be found by using the Boolean product of the matrices

$$M_{R \circ S} = M_R \cdot M_S = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

(30)

Ex: Let $A = \{1, 2, 3\}$ and $B = \{a, b, c, d\}$. Let R and S be the relations from A to B with the Boolean matrices

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \quad M_S = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

- (a) Find Boolean matrices for R^{-1} and S^{-1}
 (b) Find Boolean matrices for $(R \cap S) \circ R^{-1}$ & $R \circ R^{-1} \cap S \circ R^{-1}$
 (c) Find Boolean matrices for $S \circ (R^{-1} \cup S^{-1})$ and $S \circ R^{-1} \cup S \circ S^{-1}$

Sol: (a) Boolean matrices for R^{-1} and S^{-1} are

$$M_{R^{-1}} = (M_R)^T \\ = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M_{S^{-1}} = (M_S)^T \\ = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

(b) $M_{R \cap S} = M_R \wedge M_S$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \wedge \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{(R \cap S) \circ R^{-1}} = M_{R \cap S} \cdot M_{R^{-1}} \\ = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Now $M_{R \circ R^{-1}} = M_R \cdot M_{R^{-1}}$

$$= \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$M_{S \circ R^{-1}} = M_S \cdot M_{R^{-1}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\therefore M_{ROR'} \wedge M_{SOR'} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \wedge \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(c) \quad M_{R'} \vee M_{S'} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \vee \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$M_{SO(R'US')} = M_S \cdot M_{R'US'}$$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Again

$$M_{SOR'} = M_S \cdot M_{R'} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

And

$$M_{SOS'} = M_S \cdot M_{S'} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Now

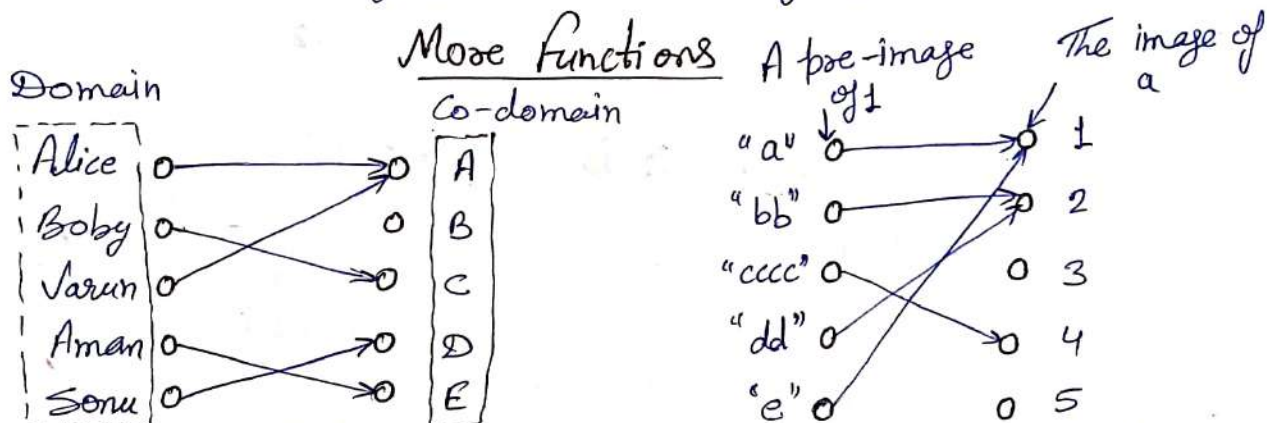
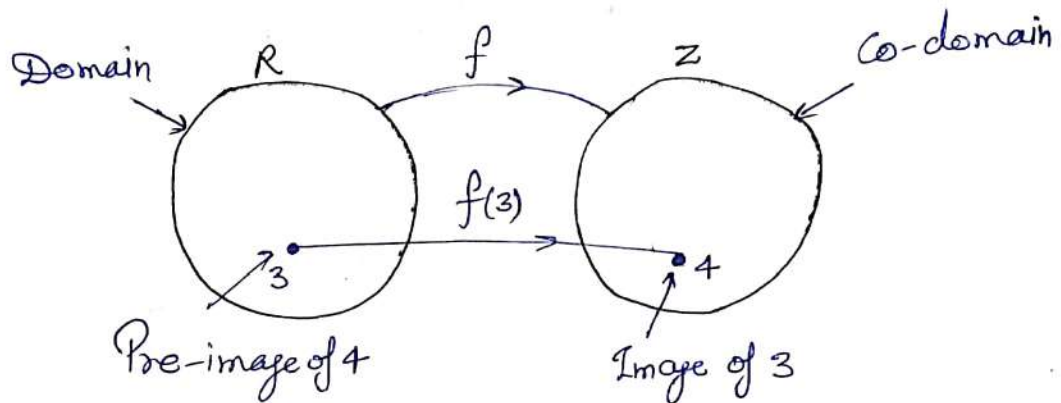
$$M_{SOR'} \vee M_{SOS'} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \vee \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

FUNCTIONS

Def: A function takes an element from a set and maps it to a UNIQUE element in another set.

* Suppose that to each element of a set A we assign a unique element of a set B; the collection of such assignments is called a Function from A into B.

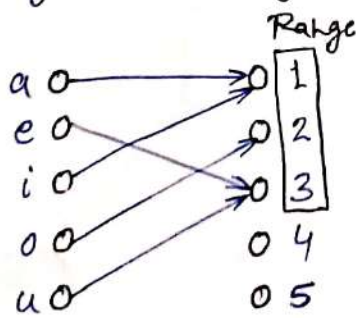
Define the function: $f: A \rightarrow B$



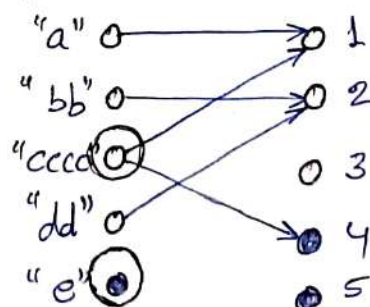
A class grade function

A string length function

NOTE: It is not necessary that each element of B shall be the image of some element of set A but it is necessary that every element of A must have one and only one image in B.



Some function



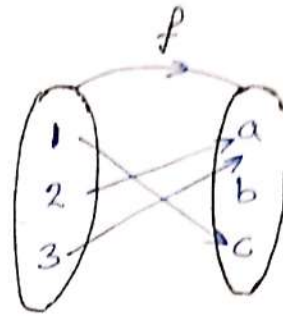
Not a valid function!

Ex.1. Let $f: \{1, 2, 3\} \rightarrow \{a, b, c\}$ defined by
 $f(1) = c, f(2) = a, f(3) = a$.

Domain = $\{1, 2, 3\}$

Co-domain = $\{a, b, c\}$

Range = $\{a, c\}$



Ex.2. Let $f: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(n) = 3n$.

Domain = $\mathbb{Z}$, Co-domain = $\mathbb{Z}$

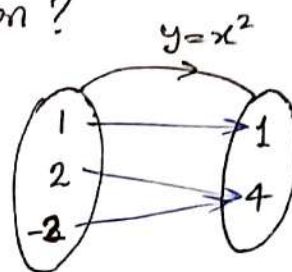
Range $(f) = 3\mathbb{Z}$

Ex.3. $y = x^2, (x \in \mathbb{R})$ Is it a function?

For $x = 1, 2, -2$

$y = 1, 4, 4$

Yes, it is a function.

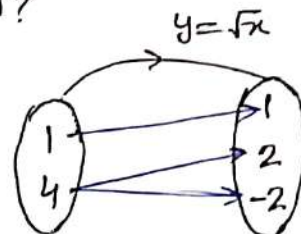


Ex.4. $y = \sqrt{x}, (x \in \mathbb{R})$ Is it a function?

For $x = 1, 4$

$y = 1, 2, -2$

No, it is not a function.



Function Arithmetic

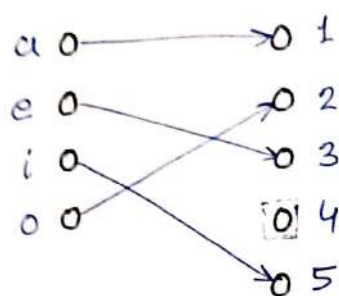
Let $f_1(x) = 2x, f_2(x) = x^2$

then $f_1 + f_2 = (f_1 + f_2)(x) = f_1(x) + f_2(x) = 2x + x^2$

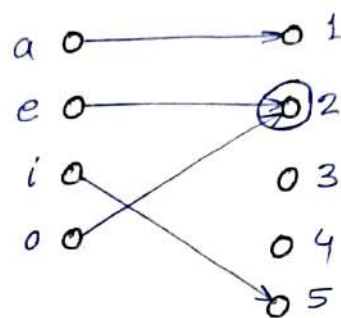
and $f_1 \times f_2 = (f_1 \times f_2)(x) = f_1(x) \times f_2(x) = 2x \times x^2 = 2x^3$

One-to-one functions

A function is one-to-one if each element in the co-domain has a unique pre-image.



A one-to-one function



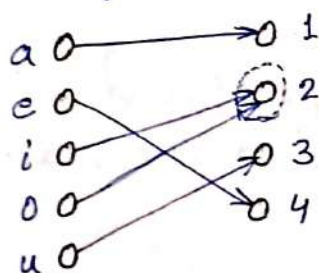
A function that is not one-to-one.

* A function is an injection if it is one-to-one.

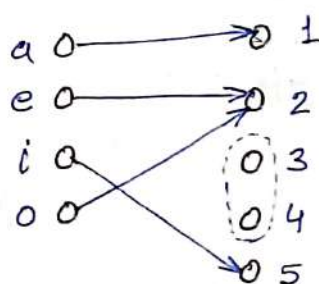
* Note that there can be un-used elements in the co-domain.

Onto Functions

A function is onto if each element in the co-domain is an image of some pre-image.



An onto function

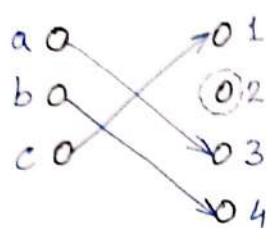


A function that is not onto.

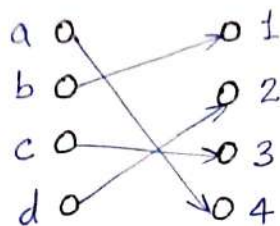
* A function is a surjection if it is onto.

* Note that there can be multiply used elements in the co-domain.

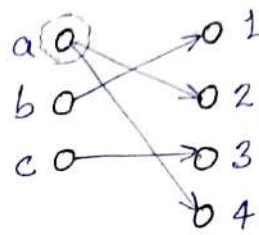
Ex. Are the following functions onto, one-to-one, both or neither?



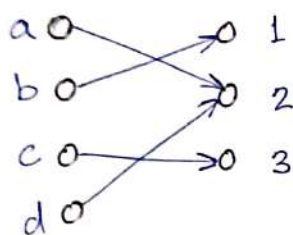
1-to-1, not onto



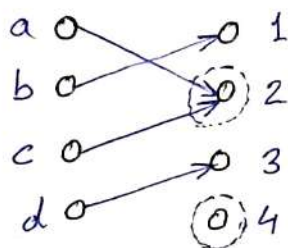
Both 1-to-1 & onto



Not a valid f^n



Onto, not 1-to-1



Neither 1-to-1 nor onto

Bijections

A function that is both one-to-one and onto is called Bijection.

Equal Functions:

Consider two functions f and g from the set A to B . The two functions f and g are said to be equal if & only if $f(a) = g(a) \quad \forall a \in A$

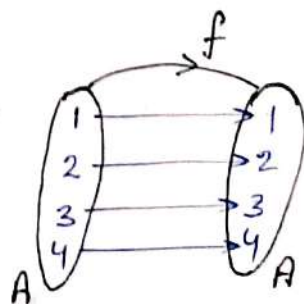
NOTE: We say two functions f and g are equal if they have the same co-domain and the same domain.

Unequal Functions:

If there exists at least one element a in A such that $f(a) \neq g(a)$, then f and g are called unequal functions.

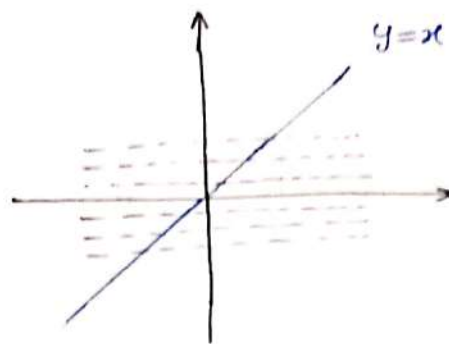
Identity Function: Let us consider a set A .

A mapping $f: A \rightarrow A$ is called the identity function, i.e., each element of A is mapped to itself.

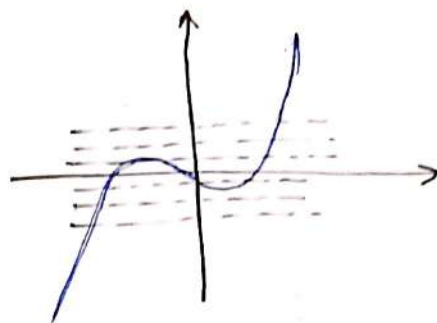


Graphical Representation of Functions

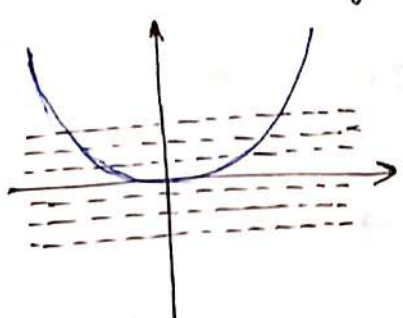
One-One: Graphically we can say that a function is one-one if each horizontal line intersects the graph of f in at most one point.



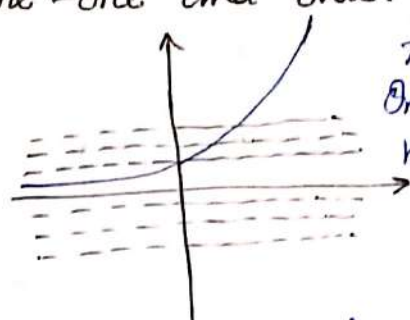
Onto: A function is onto if each horizontal line intersects the graph of f at one or more points.



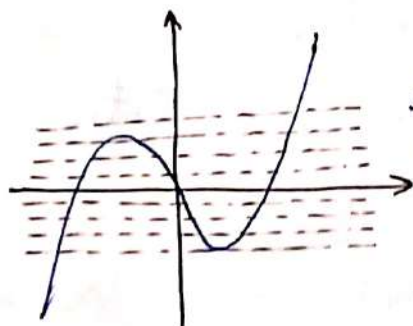
Ex. Check the functions are one-one and onto.



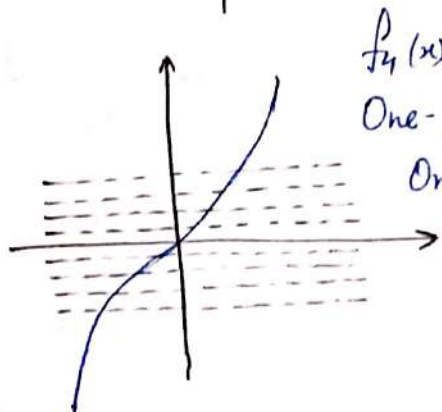
$f_1(x)$
Neither one-one
nor onto



$f_2(x)$
One-One
not Onto



$f_3(x)$
Not one-one
but onto



$f_4(x)$
One-One,
Onto

Methods to check one-one and Onto

One-One: A function $f: A \rightarrow B$ is one-one or injective, if for all elements x_1, x_2 in A such that

$$f(x_1) = f(x_2)$$

$$\Rightarrow x_1 = x_2$$

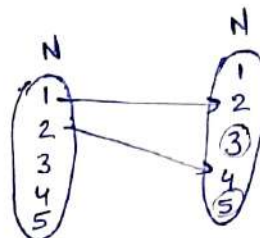
Onto: To check whether $y = f(x)$ from a set A to B is onto or not, write x in terms of y and see if for every $y \in B$, $x \in A$. If so, it is onto otherwise not.

Ex. Check the function $f: \mathbb{N} \rightarrow \mathbb{N}$, $f(x) = 2x$ is one-one & onto.

Sol. One-One: Let $f(x_1) = f(x_2)$

$$\Rightarrow 2x_1 = 2x_2$$

$$\Rightarrow x_1 = x_2$$



Onto: Let $y = f(x)$

$$y = 2x \Rightarrow x = y/2$$

$$x \in \mathbb{N}$$

$$f(1/2) = 2 \times \frac{1}{2} = 1$$

$$\text{but } \frac{y}{2} \notin \mathbb{N}$$

$$\Rightarrow \frac{1}{2} \notin \mathbb{N}$$

$\Rightarrow f$ is not onto.

Ex. Check the function $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 1/x$ is one-one & onto.

Sol. One-One: Let $f(x_1) = f(x_2)$

$$\Rightarrow \frac{1}{x_1} = \frac{1}{x_2} \Rightarrow x_1 = x_2$$

Onto: Let $y = f(x)$

$$y = \frac{1}{x} \text{ or } x = \frac{1}{y}$$

$$x \in \mathbb{R} \text{ \& } \frac{1}{y} \in \mathbb{R} \Rightarrow f \text{ is onto.}$$

If I say that $f: \mathbb{N} \rightarrow \mathbb{R}$

It's one-one.

$$f(1/2) = \frac{1}{1/2} = 2$$

$$\text{but } \frac{1}{2} \notin \mathbb{N} \Rightarrow f \text{ is not onto.}$$

Ex. Let $f: \mathbb{R} \rightarrow \mathbb{R}$, such that $f(x) = 2x^2 + 3$.

Check for one-one and onto.

Sol. One-One: Let $f(x_1) = f(x_2)$

$$2x_1^2 + 3 = 2x_2^2 + 3$$

$$x_1^2 - x_2^2 = 0$$

$$(x_1 - x_2)(x_1 + x_2) = 0$$

$$\Rightarrow x_1 = x_2 \text{ and } x_1 = -x_2$$

Not one-one

Onto: Let $y = f(x)$

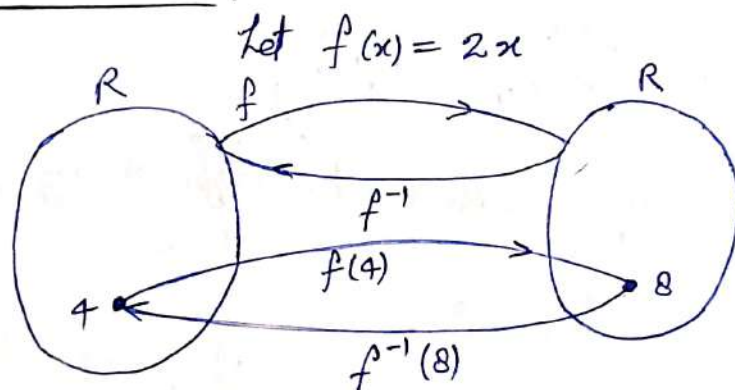
$$y = 2x^2 + 3$$

$$x^2 = \frac{y-3}{2} \Rightarrow x = \sqrt{\frac{y-3}{2}}$$

Once $y < 3$, we get imaginary number, not real.

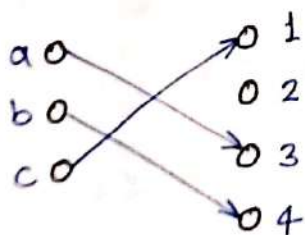
Hence f is not onto.

Inverse Functions:



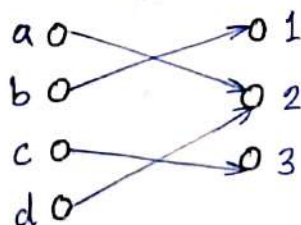
$$\text{Then } f^{-1}(x) = x/2$$

* Can we define the inverse of the following functions?



What is $f^{-1}(2)$?

Not onto!



What is $f^{-1}(2)$?

Not 1-to-1!

* An inverse function can ONLY be defined on a bijection.

Ex. Show that the mapping $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = ax + b$, where $a, b, x \in \mathbb{R}$, $a \neq 0$ is invertible. Define its inverse.

Sol For, if $x_1, x_2 \in \mathbb{R}$, then
$$f(x_1) = f(x_2) \Rightarrow ax_1 + b = ax_2 + b$$
$$ax_1 = ax_2 \Rightarrow x_1 = x_2$$

f is One-One.

$$\text{If } y \in \mathbb{R}, y \in f(x) \Rightarrow y = ax + b$$
$$\Rightarrow x = \frac{y-b}{a}$$

for $x \in \mathbb{R}$, $\frac{y-b}{a} \in \mathbb{R}$

f is onto

$$f\left(\frac{y-b}{a}\right) = a\left(\frac{y-b}{a}\right) + b = y - b + b = y$$

therefore f^{-1} exist.

$$f^{-1}(y) = \left(\frac{y-b}{a}\right)$$

Ex. If the mapping $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$, find $f^{-1}(x)$.

Sol. let $y = f(x) = x^2$
 $x = \pm\sqrt{y}$

$x \in \mathbb{R}$ but $\sqrt{y} \notin \mathbb{R}$

So f is not onto.

Thus, $f^{-1}(x)$ does not exist.

if $y = -3$

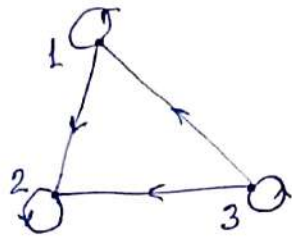
then $\sqrt{-3} = \sqrt{3}i$ is not real.

Partial Ordered Set (POSET)

①

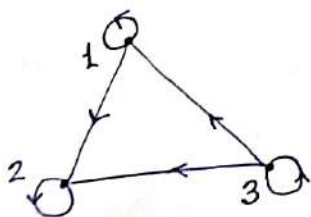
A relation R on a set S is called a partial ordering or partial order if it is reflexive, antisymmetric and transitive. A set S together with a partial ordering R is called a partially ordered set or poset, and is denoted by (S, R) .

Ex 1. let $S = \{1, 2, 3\}$ and
let $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (3, 1), (3, 2)\}$



In a poset the notation $a \leq b$ denotes that $(a, b) \in R$. The symbol $\leq$ is used to denote the relation in any poset, not just the "less than or equals" relation. The notation $a < b$ denotes that $a \leq b$, but $a \neq b$.

Ex. let $S = \{1, 2, 3\}$ and
let $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (3, 1), (3, 2)\}$



$$2 \leq 2, \quad 3 \leq 2.$$

if I say $a \leq b$, we say a precedes b and b succeeds a .
 3 strictly precedes 2 and 2 strictly succeeds 3 .

Ex 2. Show that $\geq$ is a partial order on the set of integers.

- It is reflexive: $a \geq a$ for all $a \in \mathbb{Z}$.
- It is antisymmetric: if $a \geq b$ then the only way that $b \geq a$ is when $b = a$.
- It is transitive: If $a \geq b$ & $b \geq c$, then $a \geq c$.

②

Note that $\geq$ is the partial ordering on the set of integers.
 $(\mathbb{Z}, \geq)$ is the partially ordered set, or poset.

Ex.3 Consider the power set of $\{a, b, c\}$ and the subset relation. $(P(\{a, b, c\}), \subseteq)$

Draw a graph of this relation.

$$P(\{a, b, c\}) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{c, a\}, \{a, b, c\}\}$$

Every set is its own subset $\Rightarrow$ Reflexive

$$A \subseteq B \text{ \& } B \subseteq A \Rightarrow A = B$$

So, Antisymmetric.

$$A \subseteq B \text{ \& } B \subseteq C \Rightarrow \underline{A \subseteq C} \text{ so transitive.}$$

Thus, $(P(\{a, b, c\}), \subseteq)$ is a partial order relation or Poset.

Partial Ordering:

Suppose a binary relation $\geq$ contains the pairs (a, b) as defined in the following cases

In which cases will $\geq$ be a partial ordering?

1. (a, b) such that $a \leq b$; $a, b \in \mathbb{R}$ TRUE
2. (a, b) such that $a < b$; $a, b \in \mathbb{R}$ FALSE
3. (a, b) such that $(a-b)$ is divisible by 3; $a, b \in \mathbb{Z}$ FALSE
 $3-6 = -3$ \& $6-3 = 3$ but $3 \neq 6$.
4. (a, b) such that a divides b ; $a, b \in \mathbb{Z}^+$ TRUE

Ex. Let S be a set of people.

Suppose a binary relation $\leq$ contains the pairs as defined in the following cases

In which cases will $\leq$ be a partial ordering?

1. (A, B) such that $A \subseteq B$; $A, B \in 2^S$ ^{Power Set} ~~TRUE~~

2. (x, y) such that x is older than y ; $x, y \in S$ FALSE

3. (x, y) such that x is not older than y ; $x, y \in S$ TRUE
 $x \neq y \Rightarrow x \leq y$

Comparable / Incomparable.

The elements a and b of a poset $(S, \leq)$ are called comparable if either $a \leq b$ or $b \leq a$.

When a and b are elements of S such that neither $a \leq b$ nor $b \leq a$, a and b are called incomparable.

Ex. Consider the power set of $\{a, b, c\}$ and the subset relation. $(P(\{a, b, c\}), \subseteq)$.

$\{a, c\} \not\subseteq \{a, b\}$ and $\{a, b\} \not\subseteq \{a, c\}$

So, $\{a, c\}$ and $\{a, b\}$ are incomparable.

whereas, $\{a\} \subseteq \{a, b\}$ are comparable.

Totally Ordered, chains

If $(S, \leq)$ is a poset and every two elements of S are comparable, S is called totally ordered or linearly ordered set, and $\leq$ is called a total order or a linear order. A totally ordered set is also called a chain.

Ex.

In the poset $(\mathbb{Z}^+, \leq)$, are the integers 3 and 9 comparable?

• Yes, as $3 \leq 9$.

(4)

- Are 7 and 5 comparable?
- Yes, as $5 \leq 7$
- As all pairs of elements in $\mathbb{Z}^+$ are comparable, the poset $(\mathbb{Z}^+, \leq)$ is a total order.
- So it is totally ordered poset, linear order, or chain.

Ex.

- In the poset $(\mathbb{Z}^+, |)$ with "divides" operator $|$, are the integers 3 and 9 comparable?
- Yes, as $3|9$
- Are 7 and 5 comparable?
- No, as $7 \nmid 5$ and $5 \nmid 7$
- Thus, as there are pairs of elements in $\mathbb{Z}^+$ that are not comparable, the poset $(\mathbb{Z}^+, |)$ is a partial order. It is not a chain.

Hasse Diagrams

Given any partial order relation defined on a finite set, it is possible to draw the directed graph so that all of these properties are satisfied.

This makes it possible to associate a somewhat simpler graph, called a Hasse diagram, with a partial order relation defined on a finite set.

* Since partial ordering is a binary relation, it can be represented by a directed graph.

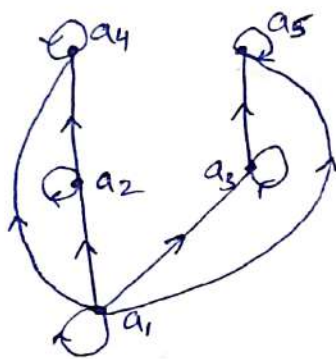
* However, many edges can be omitted, because such an ordering must be reflexive and transitive.

* Also, we may order the vertices in the graph in a 'vertical' manner, such that all edges are pointing from low to high
 → directions on an edge can be omitted.

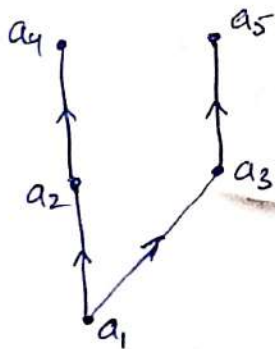
1. Remove all the self loops.
2. Remove all the edges that must be present due to transitivity.
3. Arrange all edges to point upwards
4. Do not show directions on the edges.

Ex. Let $A = \{a_1, a_2, a_3, a_4, a_5\}$

$$R = \{(a_1, a_1), (a_2, a_2), (a_3, a_3), (a_4, a_4), (a_5, a_5), (a_1, a_2), (a_1, a_3), (a_1, a_4), (a_1, a_5), (a_2, a_4), (a_3, a_5)\}$$



Remove Self-loops



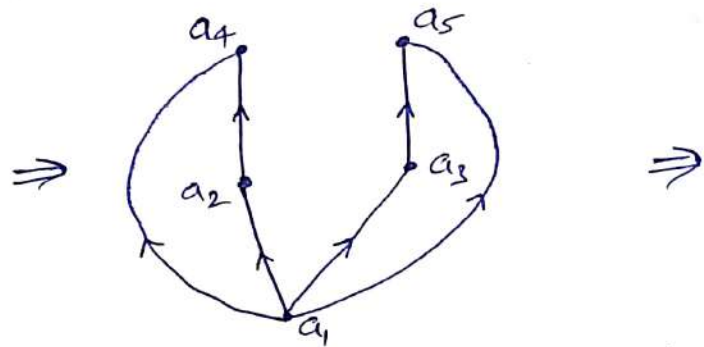
Remove Directions

If I talk about $a_1 \leq b \nexists c \mid a_1 \leq c \& c \leq b$

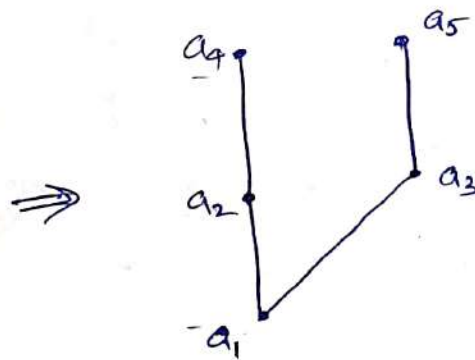
Here, $a_1 \leq a_2$, $a_1 \leq a_3$, $a_2 \leq a_4$, $a_3 \leq a_5$

We cannot say that $a_1 \leq a_4$ since a_2 between them.

$$a_2 \not\leq a_3$$



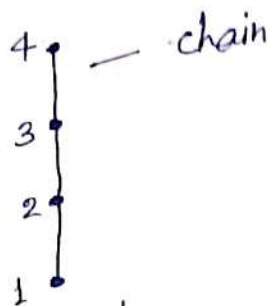
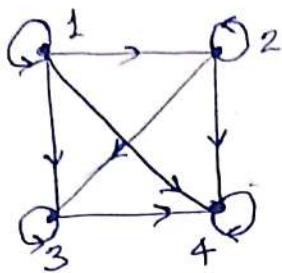
Remove Transitive Loops



Hasse Diagram

$$\begin{aligned} a_1 &\leq a_2 \\ a_1 &\leq a_3 \\ a_2 &\leq a_4 \\ a_3 &\leq a_5 \end{aligned}$$

⑥ Ex. Consider the poset $(\{1, 2, 3, 4\}, \leq)$



Hasse diagram

It's not necessary that you always start with the digraph and then you make the Hasse diagram. You can draw the Hasse diagram from sketch also.

Ex. Draw a Hasse diagram for the partial ordering $\{(a, b) : a|b\}$

on $\{1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60\}$ or D_{60}

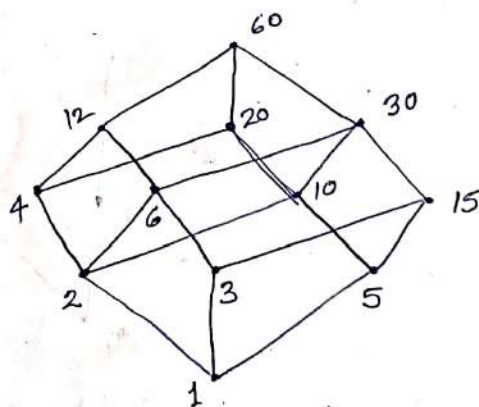
Sol. $1 < 2, 1 < 3, 1 < 4$ but $1 \nless 4$ since $1 < 2 < 4$

$2 < 6$ since

$2 < 4 \nless 6$

$a < b \nless c$

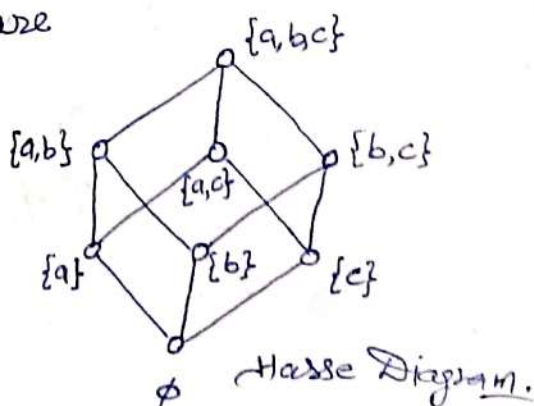
$a < c$ & $c < b$



Ex. Construct the Hasse diagram of $(P(\{a, b, c\}), \subseteq)$

Sol. The elements of $P(\{a, b, c\})$ are

$\emptyset$
 $\{a\} \quad \{b\} \quad \{c\}$
 $\{a, b\}, \{a, c\}, \{b, c\}$
 $\{a, b, c\}$



Maximal and Minimal Elements

Def: An element a in a poset $(S, \leq)$ is called maximal if it is not less than any other element in S . That is,

$$\nexists b \in S (a < b).$$

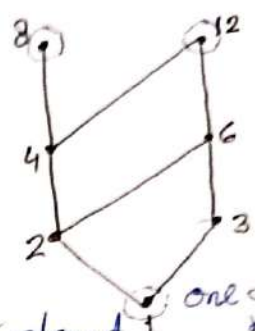
An element which is not preceded by any other element. one or more

Def: An element a in a poset $(S, \leq)$ is called minimal if it is not greater than any other element in S .

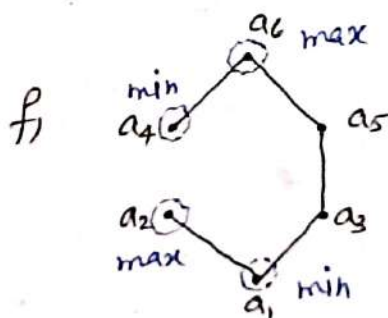
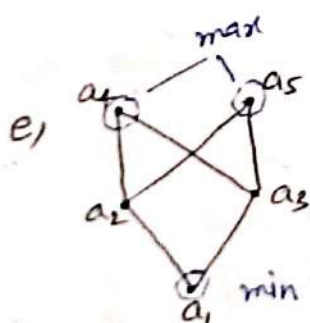
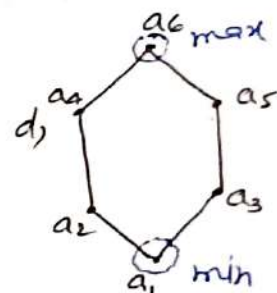
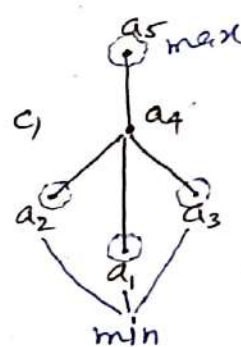
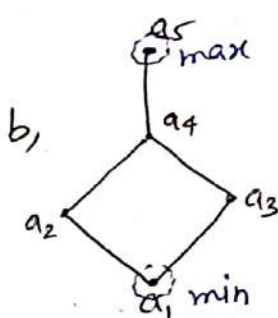
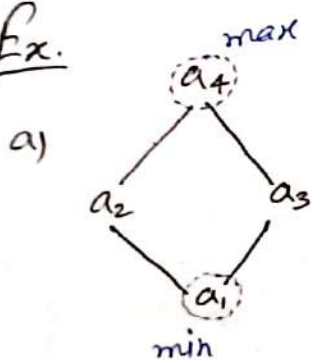
That is,

$$\nexists b \in S (b < a)$$

An element which is not succeeded by any other element. May be one or more.



Ex.



Greatest Element:

a is the greatest element in the poset $(S, \leq)$ if $b \leq a$ for all $b \in S$.

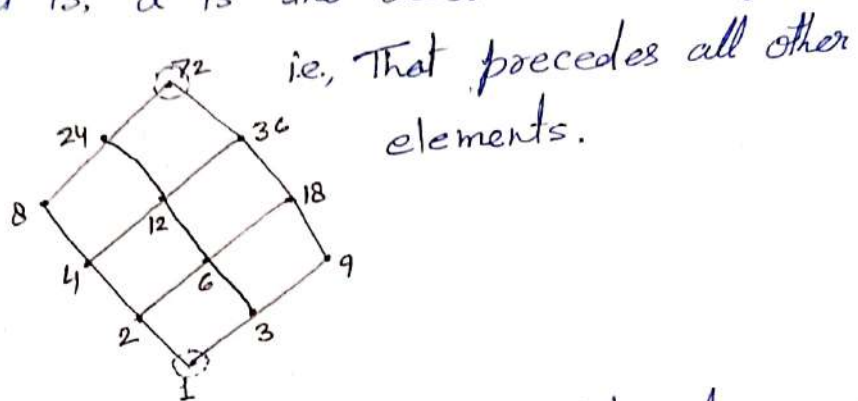
OR

Greatest element is that succeeds all other elements.

(A)

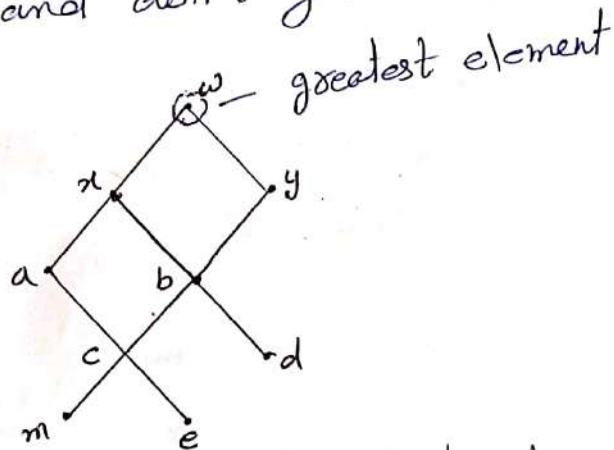
Least Element:

An element of a poset is called the least element if it is less or equal than all other elements in the poset. That is, a is the least element if $a \leq b \forall b \in S$.



Greatest and least elements are optional in nature. There might be cases that you do not get a greatest element and don't get a least element.

Ex.



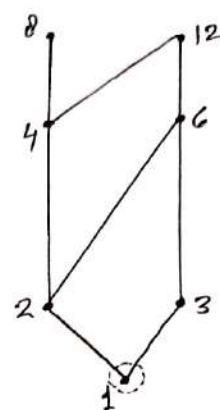
Here, w is the greatest element. But what about the least element?

If I say that m is the least element but m does not precede e . If I say e is the least element e doesn't precede m or d . So in this particular case, I have a greatest element but I don't have a least element.

A Simple Thumb Rule is: If you have multiple minimal elements, you will not have a least element.

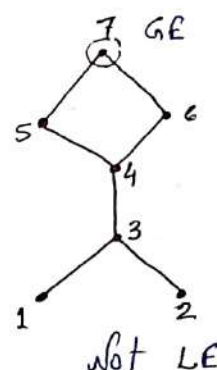
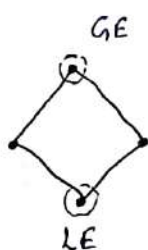
Similarly, let's see this case:

Here I can very well see that I have a least element 1. But Out of these two (8, 12), I can't say which one is the greatest element, because I have multiple maximal elements.



Greatest and least elements will always exist iff no. of maximal elements = 1 and minimal elements = 1 respectively.

Ex.



Upper Bound:

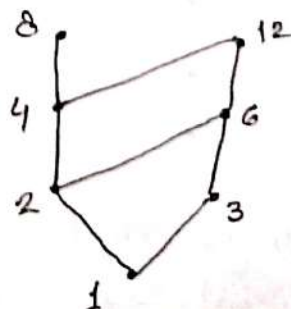
Let $(S, \leq)$ be a poset and let $A \subseteq S$. If u is an element of S such that $a \leq u$ for all elements $a \in A$ then u is an upper bound of A .

Lower Bound:

Let $(S, \leq)$ be a poset and let $A \subseteq S$. If l is an element of S such that $l \leq a$ for all elements $a \in A$ then l is a lower bound of A .

Ex Which items are:

1. Upper bounds of 2 and 3?
2. Lower bounds of 2 & 3?
3. Upper bounds of 6 & 8?
4. Lower bounds of 6 & 8?



Upper bound of this subset $\{2, 3\}$ is the set of all the elements from this poset which succeed both 2 and 3.

The elements that are succeeded 2

$$2 \Rightarrow 2, 6, 4, 8, 12$$

So all the elements which can be reached from 2, including 2 by going in the upward direction, i.e., the upper bound of element 2.

$$UB \{2\} = \{2, 6, 4, 8, 12\}$$

$$UB \{3\} = \{3, 6, 12\}$$

$$UB \{2, 3\} = \{6, 12\}$$

Now. $UB \{6\} = \{6, 12\}$, $UB \{8\} = \{8\}$

$$UB \{6, 8\} = \emptyset$$

For lower bound we will check the elements which are preceding the given element.

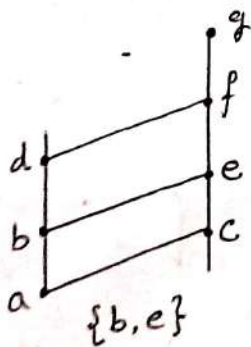
$$LB \{2\} = \{1\} , LB \{3\} = \{1, 3\}$$

$$LB \{2, 3\} = \{1\}$$

$$LB \{6\} = \{6, 2, 3, 1\} , LB \{8\} = \{8, 4, 2, 1\}$$

$$LB \{6, 8\} = \{2, 1\}$$

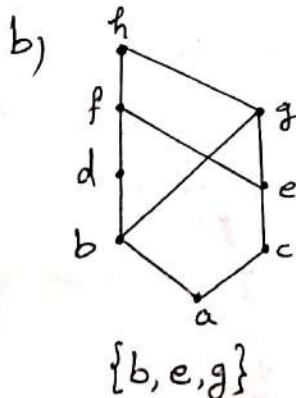
Ex.
a)



$$UP \{b\} = \{b, d, e, f, g\}$$

$$UB \{e\} = \{d, f, e, g\}$$

$$UB \{b, e\} = \{e, f, g\}$$

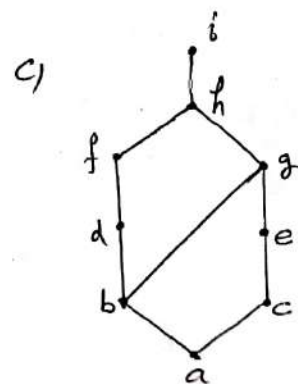


$$\{b, e, g\}$$

$$UB \{b\} = \{b, d, f, h, g\}$$

$$UB \{e\} = \{f, g, h\}$$

$$UB \{g\} = \{g, h\}$$



$$\{b, d\}$$

$$UB \{b\} = \{b, d, f, h, i, g\}$$

$$UB \{d\} = \{f, d, h, i\}$$

$$UB \{b, d\} = \{d, f, h, i\}$$

$$LB\{b\} = \{b, a\}$$

$$LB\{e\} = \{e, b, c, a\}$$

$$LB\{b, c\} = \{b, a\}$$

$$UB\{b, e, g\} = \{g, h\}$$

$$LB\{b\} = \{b, a\}$$

$$LB\{e\} = \{e, c, a\}$$

$$LB\{g\} = \{g, e, c, a, b\}$$

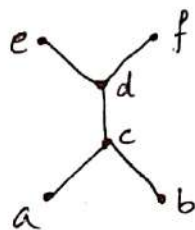
$$LB\{b, e, g\} = \{a\}$$

$$LB\{b\} = \{b, a\}$$

$$LB\{d\} = \{d, b, a\}$$

$$LB\{b, d\} = \{b, a\}$$

Ex.



$$LB\{a, b\} ?$$

$$LB\{a\} = \{a\}, \quad LB\{b\} = \{b\}$$

$$LB\{a, b\} = \emptyset$$

$$UB\{e, f\} ?$$

$$UB\{e\} = \{e\}, \quad UB\{f\} = \{f\}, \quad UB\{e, f\} = \emptyset$$

So it's not necessary that for every subset, you will get a upper bound or a lower bound, upper and lower bound might be null also.

Least Upper Bound:

An element x that is an upper bound on a subset A and is less than all other upper bounds on A is called the least upper bound on A . We abbreviate "lub". Also called the Supremum.

Greatest Lower Bound:

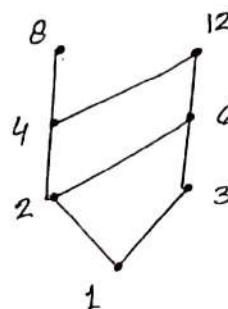
An element x that is a lower bound on a subset A and is greater than all other lower bounds on A is called the greatest lower bound on A . We abbreviate "glb". Also called the infimum.

NOTE: We denote $\text{lub}(\{a, b\})$ by $a \vee b$ and call it joint or sum of a and b . Similarly,

We denote $\text{glb}(\{a, b\})$ by $a \wedge b$ and call it meet or product of a & b .

Ex. Which items are:

1. LUB of 2 and 3?
2. GLB of 2 and 3?
3. LUB of 6 and 8?
4. GLB of 6 and 8?



Sol. 1. $UB\{2, 3\} = \{6, 12\}$

$LUB = \{6\}$

2. $LB\{2, 3\} = \{1\}$ so $GLB = \{1\}$

3. $UB\{6, 8\} = \emptyset$ [$\because UB\{6\} = \{6, 12\}$ & $UB\{8\} = \{8\}$]

$LUB = \emptyset$

4. $LB\{6, 8\} = \{2, 1\}$

$GLB = \{2\}$

Ex. Consider the poset $A = \{a, b, c, d, e, f, g\}$ be ordered shown in Fig. Also let $B = \{c, d, e\}$. Determine the upper & lower bound of B .

Sol. Succeeding by $c \Rightarrow c, e, f, g$
 $d \Rightarrow d, e, f, g$
 $e \Rightarrow e, f, g$

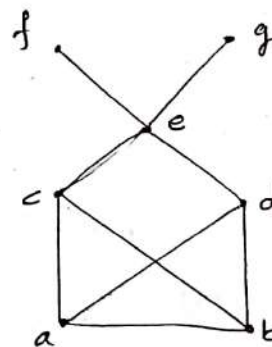
$UB\{c, d, e\} = \{e, f, g\}$

$LUB = \{e\}$

Preceded by $c \Rightarrow c, b, a$
 $d \Rightarrow d, b, a$
 $e \Rightarrow e, c, d, b, a$

$LB = \{b, a\}$

$GLB = \emptyset$



Ex. Determine the least upper bound and greatest lower bound of $B = \{a, b, c\}$ if they exist, of the poset whose Hasse diagram is shown in fig:

Sol. Succeeded by $a \Rightarrow a, c, d, e$
 $b \Rightarrow b, c, d, e$
 $c \Rightarrow c, d, e$

$$UB = \{c, d, e\}$$

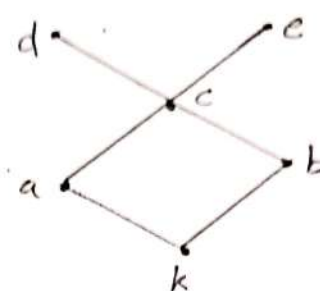
$$LUB = \{c\}$$

Preceded by $a \Rightarrow a, k$

$$b \Rightarrow b, k$$

$$c \Rightarrow c, a, b, k$$

$$LB = \{k\} \text{ So } GLB = \{k\}$$



Ex. What are the lower/upper bounds and glb/lub of the sets $\{d, e, f\}$, $\{a, c\}$ and $\{b, d\}$.

Sol. $\{d, e, f\}$

Preceded by $d \Rightarrow d, b$

$$e \Rightarrow e$$

$$f \Rightarrow f, b$$

So, $LB = \emptyset$, thus no glb either.

Succeeded by $d \Rightarrow d, g$

$$e \Rightarrow e, g, i$$

$$f \Rightarrow f, i$$

• $UB = \emptyset$, thus no lub either.

$\{a, c\}$

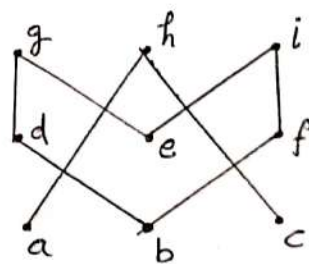
• Lower Bounds: $\emptyset$, thus no glb either.

• Upper Bounds: $\{h\}$, since its unique, lub is also h.

$\{b, d\}$

• Lower Bounds: $\{b\}$ and so also glb is b.

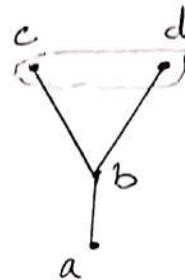
• Upper Bounds: $\{d, g\}$ and since $d \leq g$, the lub is d.



LATTICES

Def. A partially ordered set in which every pair of elements has both a least upper bound and a greatest lower bound is called a Lattice.

Ex. Is the shown Hasse Diagram a Lattice?



Can you find the LUB (Supremum)

of $\{c, d\}$? ϕ

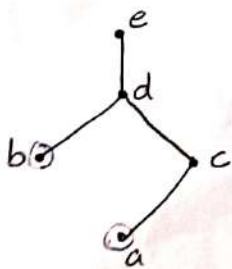
So if I can find a pair of elements which does not have a Supremum then I can say that it is not a Lattice.

So in this case, it is not a Lattice.

We can say that,

NOTE: A Hasse Diagram that does not have a clearly defined Greatest/and/or Least element cannot be a Lattice.

Ex



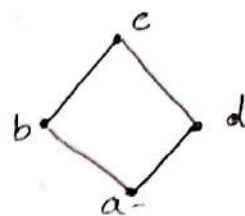
Here both a & b are minimal elements and if there are two minimal elements that means there is no clearly defined least element.

If I take the pair $\{a, b\}$ over here. This pair will not have any lower bound. So it does not have a lower bound that means a infimum does not exist. So it is not a Lattice.

Ex of Lattices

How can I say that it is a Lattice?

I will take all the possible pairs and find out the Infimums and Supremums of all the pairs and if I can find out all the Infimums and Supremums then this is a Lattice.



We can find out the

Supremum (LUB)

v	a	b	c	d
a	a	b	c	d
b	b	b	c	c
c	c	c	c	c
d	d	c	c	d

Infimum (GLB)

^	a	b	c	d
a	a	a	a	a
b	a	b	b	a
c	a	b	c	d
d	a	a	d	d

The Supremum of $a \vee a$ is a , $a \vee b$ is b .

You can find out of Supremum over here. So, $a \vee a$ the upper bound of $a = \{a, b, c, d\}$

$$\text{LUB} = \{a\}$$

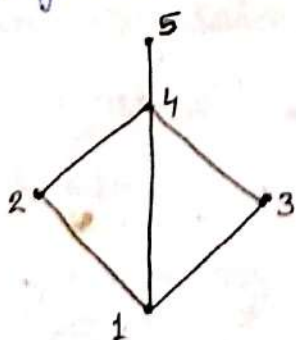
$$\text{LB } \{a, a\} = \{a\}$$

Similarly, you can find out the Supremum and Infimum of all the elements. Now you see all the Supremums exist.

There is no pair that is missing a Infimum or a Supremum. As I can find out all the, how many elements are there a, b, c, d . 4 so that means there should be 16 Supremums and 16

Infimums. So, if I can find out all the 16 Supremums and all 16 Infimums that means it is a Lattice.

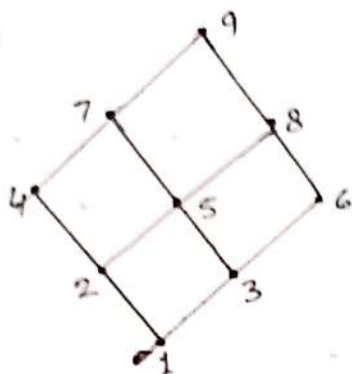
Ex



v	1	2	3	4	5
1	1	2	3	4	5
2	2	2	4	4	5
3	3	4	3	4	5
4	4	4	4	4	5
5	5	5	5	5	5

^	1	2	3	4	5
1	1	1	1	1	1
2	1	2	1	2	2
3	1	1	3	3	3
4	1	2	3	4	4
5	1	2	3	4	5

Ex.



As the cardinality of Poset increases, it becomes difficult to make the table. Can we get some shortcuts?

Let us see, try to build the tables of Infimum and Supremum of this diagram by using certain shortcuts.

Supremum

	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	2	5	4	5	8	7	8	9
3	3	5	3	7	5	6	7	8	9
4	4	4	7	4	7	9	7	9	9
5	5	5	5	7	5	8	7	8	9
6	6	8	6	9	8	6	9	8	9
7	7	7	7	7	7	9	7	9	9
8	8	8	8	9	8	8	9	8	9
9	9	9	9	9	9	9	9	9	9

Infimum

	1	2	3	4	5	6	7	8	9
1	1	1	1	1	1	1	1	1	1
2	1	2	1	2	2	1	2	2	2
3	1	1	3	1	3	3	3	3	3
4	1	2	1	4	2	1	4	2	4
5	1	2	3	2	5	2	5	5	5
6	1	1	3	1	3	6	3	6	6
7	1	2	3	4	5	3	7	5	7
8	1	2	3	2	5	6	5	8	8
9	1	2	3	4	5	6	7	8	9

First of all please remember two operations join and meet operations. The join ($\vee$) operation representing the Supremum and meet ($\wedge$) representing the Infimum.

The first clue or the first shortcut is that both the join and meet operations are Idempotent.

$$\underline{a \vee a = a} \quad \text{and} \quad \underline{a \wedge a = a}$$

So by using this simple property what you can do? You can fillup without any checking the Supremum and Infimum, you can simply fill in all the diagonal elements.

Now the second rule says

If $a \leq b$ then $a \vee b = b$ and $a \wedge b = a$

By using this property we can fill in this

As 1 is preceding all the other elements so the join of 1 & 2 is 2. Similarly the meet $1 \wedge 2 = 1$.

Third property and very important property is that both the join and meet operations are commutative in nature, i.e.,

$$\underline{a \vee b = b \vee a} \quad \text{and} \quad \underline{a \wedge b = b \wedge a}$$

That means every row is going to mimic its corresponding column.

I will use these properties repeatedly now. See the element 2, element 2 precedes by 4, 5, 7, 8, 9, So the join of 2 & 4 will be 4, 5, 7, 8, 9. Similarly the meet will be 2, 2, 2, 2, 2 and as operation is commutative, so I can fill in the columns also. Next take the element 3, element 3 precedes 3, 5, 6, 7, 8, 9. So I can put in. then again 3, 3, Again commutative in nature. Next take the element 4, --- and so on.

Now the remaining cells are the one's where I actually have to find the glb and lub. Out of 81 elements that I needed to find only the non-comparable pairs are left. See 2 & 3 are non-comparable, so 2, 3 pair is left, 3, 2 pair is left.

So out of 81 only 18 pairs are remaining. So the task has become much easier. Now what we are going to do. We are going to find the remaining Infimums and Supremums.

Now once again for that also, I am going to give you a simple trick. I have to find out the supremum of 2 & 3.

For Supremum try to find out the element where the upward.

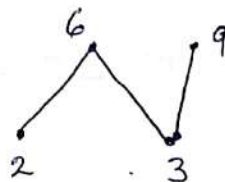
18

lines from these two elements are matching. See, from 2 there are many lines going upwards, from 3 there are once again 2 lines. Now try to find out the lowest element in the order where the lines are matching. So the supremum of 2 and 3 is going to be 5. You can verify by making the upper and lower bound but this is just a simple trick that I am taking. Similarly, if I want to find the Infimum then go to downwards and see what is the element where the lines are matching.

So all pair of elements has both infimum and Supremum, the given Hasse Diagram is a Lattice.

Examples on Hasse Diagram:

- 1, { Non-Trivial Divisors of 18 }
 { 2, 3, 6, 9 }



- 2, { Non-Trivial Divisor of 6 }
 { 2, 3 }



Sublattice:

Let $(L, \leq)$ be a lattice. A nonempty subset S of L is called a sublattice of L if $a \vee b$ in S and $a \wedge b$ in S whenever a and b in S .

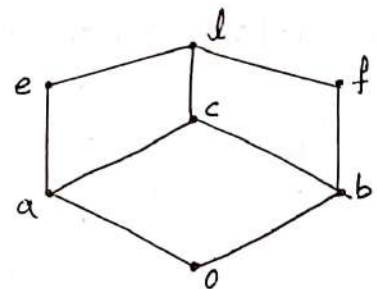
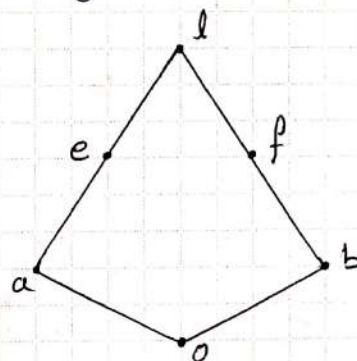
- Sublattice of a lattice must itself be a lattice.
- Every Singleton set of a lattice is a sublattice of L .
- If $m|n$ (m divides n) then D_m is a sublattice of D_n .
where $D_m = \text{Set of Positive Divisors}$

Let us say $D_{12} = \{1, 2, 3, 4, 6, 12\}$ and if I say
4 divides 12

So, D_4 sublattice of D_{12} . Will be clarify later on.

Example: Consider the lattice $(L, \leq)$ as shown below.

. Now, I am taking a subset L_1



Now, I want to verify whether L_1 is a sublattice of L or not.

So the first condition says L_1 must be a lattice itself. Now you know how to check for a lattice. You have to check for the Infimum or Supremum of each & every pair. OR if you want to expedite the task, just identify the non-comparable pairs first.

If there are 2 non-comparable pairs I can see over here, (a, b) and (e, f) but in this case if I see (a, b) , it has

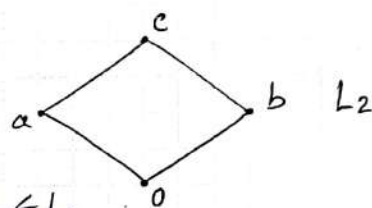
infimum of 0 and Supremum 1. (e, f) has an infimum 0 and supremum 1, so if the non-comparable elements have both Supremum and infimum so as a Thumb Rule you can say each & every pair is going to have a infimum and Supremum. So it passes the first condition, it is a Lattice.

Now the second condition is, I have to check whether every pair of elements of this subset if it is considered to be a pair of element from the Original set and we identify the infimum and supremum that must exist in the subset also.

So let's take the pair (a, b) and check it.

L_1 is not sublattice of L since $a \vee b = c \notin L_1$
 Let's take another example.

L_2 is sublattice in L since



$$a \vee 0 = a \in L_2$$

$$a \wedge 0 = 0 \in L_2$$

$$b \vee 0 = b \in L_2$$

$$b \wedge 0 = 0 \in L_2$$

$$a \vee c = c \in L_2$$

$$a \wedge c = a \in L_2$$

$$b \vee c = c \in L_2$$

$$b \wedge c = b \in L_2$$

$$0 \vee c = c \in L_2$$

$$0 \wedge c = 0 \in L_2$$

$$0 \vee 0 = 0 \in L_2$$

$$0 \wedge 0 = 0 \in L_2$$

$$a \vee a = a \in L_2$$

$$a \wedge a = a \in L_2$$

$$b \vee b = b \in L_2$$

$$b \wedge b = b \in L_2$$

$$c \vee c = c \in L_2$$

$$c \wedge c = c \in L_2$$

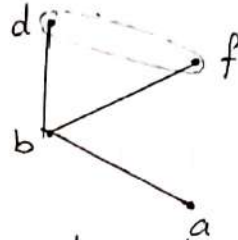
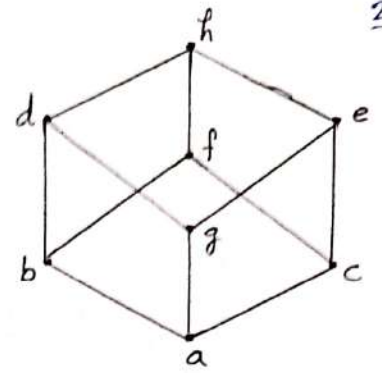
For all $a, b \in L_2$, $a \vee b \in L_2$ and $a \wedge b \in L_2$.

Hence L_2 is sublattice of L .

Example:

21

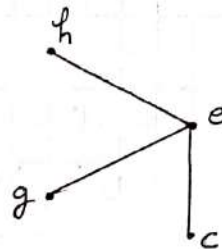
Lets use subset $L_1 = \{a, b, d, f\}$
If a subset is given like this, the easiest way to start is by drawing the subset,
lets draw the subset



It has multiple maximum elements, it cannot be a lattice.
 d & f are non-comparable elements and if you try to find out the upper bound of d & f , there is no upper bound. So d & f is not going to have a supremum, so this is not a lattice. Thus, it fails the first test. So it is not a sublattice.

In the given lattice L , $d \vee f = h \notin L$,
 L_1 is not a sublattice of L .

$$L_2 = \{c, e, g, h\}$$

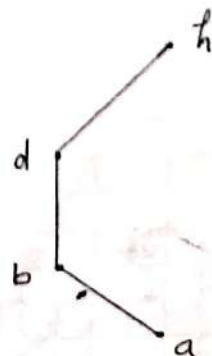


Here we have two minimal elements, ~~so~~ g & c . So obviously g & c are not going to have any infimum. Thus it cannot be a lattice. and if it is not a lattice, it cannot be a sublattice.

$$L_3 = \{a, b, d, h\}$$

Lets draw it.

Now to check, it is a lattice, form tables of Supremum and infimum of each pair of elements.



$\vee$	a	b	d	h
a	a	b	d	h
b	b	b	d	h
d	d	d	d	h
h	h	h	h	h

$\wedge$	a	b	d	h
a	a	a	a	a
b	a	b	b	b
d	a	b	d	d
h	a	b	d	h

Hence for every $a, b \in L_3$

$$a \vee b \in L_3 \text{ and } a \wedge b \in L_3$$

Hence, L_3 is sublattice of L .

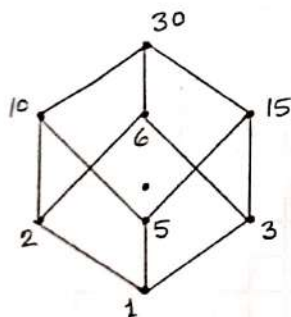
Example:

Let $D(n)$ denote the positive divisors of n .

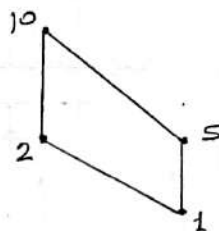
$$L = D(30) = \{1, 2, 3, 5, 6, 10, 15, 30\}$$

- For the divisor relation, if $n \div m$ then $D(m)$ is a sublattice of $D(n)$.

We have taken $D(30)$ has an example over here.



It is a Lattice, you can verify it.
And as a subset, I have taken
 $D(10) = \{1, 2, 5, 10\}$



You can see that it is a lattice itself and if you use the 4 elements to find out the infimum and Supremum, all the infimum and Supremum will come from this set $D(10) = \{1, 2, 5, 10\}$.

So this is a lattice and the join and meet both operations are closed, so this is a sublattice.

So you can verify $D(6)$, $D(10)$, $D(15)$ all of them will be sublattices.

Bounded Lattice:

A lattice L is said to be bounded if it has a greatest element 1 and a least element 0 .

Example: The lattice $P(S)$ of all subsets of a set S , with the relation containment is bounded. The greatest element is S and the least element is empty set.

Example: The lattice $\mathbb{Z}^+$ under the partial order of divisibility is not bounded, since it has a least element 1 , but no greatest element.

- If L is a bounded lattice, then for all a in A

$$0 \leq a \leq 1$$

$$a \vee 0 = a, \quad a \vee 1 = 1$$

$$a \wedge 0 = 0, \quad a \wedge 1 = a$$

Note: $1(0)$ and a are comparable, for all a in A .

Now let us see what do we actually mean by a complemented lattice. Before going into that we have to understand what is actually meant by a complement.

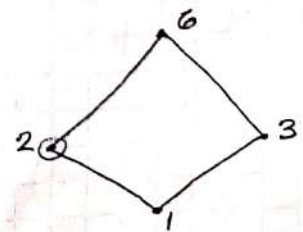
Complemented Lattice:

- If we have a Bounded lattice L with a 0 (Least) element and a 1 (Greatest) element and in this lattice, if for an element $a \in L$ we can find another element $b \in L$ such that $a \vee b = 1$ and $a \wedge b = 0$, then b is said to be the complement of a and vice-versa.

Example 1: Let an element x such that

$$2 \vee x = 6 \quad \text{and} \quad 2 \wedge x = 1$$

then x will be the complement of 2 .



Lets try to find out, lets compare 2 with 1.

$$2 \vee 1 = 2 \quad \text{and} \quad 2 \wedge 1 = 1$$

So here infimum is 1 but the Supremum is 2, thus obviously complement of 2 is not 1.

If I compare 2 & 6.

$$2 \vee 6 = 6 \quad \text{and} \quad 2 \wedge 6 = 2$$

The ~~infimum~~ supremum is 6 but the infimum is 2, so clearly the complement of 2 is not 6 also.

If I take 2 and 3,

$$2 \wedge 3 = 1 \quad \text{and} \quad 2 \vee 3 = 6$$

The infimum is 1 and supremum is 6, so I can say that the complement of 2 is 3 ($2' = 3$) and by the commutativity complement of 3 is 2. ($3' = 2$).

Now try to find out the complement of 1, so you can see that the complement of 1 cannot be 2, cannot be 3, the complement of 1 ~~can~~ is going to be 6. So

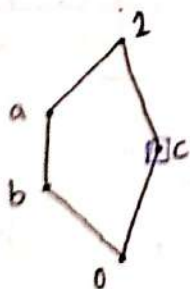
$$6' = 1 \quad \text{and} \quad 1' = 6$$

In fact, the complement of the least element = Greatest element and

$$(\text{Greatest element})' = \text{least element.}$$

- Complements are not unique, i.e., an element can have multiple complements.

Ex. 2.



If I tried to find out the complement of c.

We know that $0' = 1$ & $1' = 0$.

If I compare c with a

$$c \vee a = 1 \quad \text{and} \quad c \wedge a = 0$$

$$\text{So } c' = a$$

If I compare c with b , again the same conditions are satisfied.

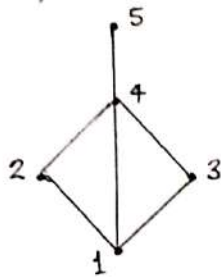
So $b' = c'$

So c has 2 complements a & b . I can say

$$a' = c \text{ \& \& } b' = c$$

- There can be elements in a lattice, which don't have any complements.

Ex. 3.



So obviously here also

$$1' = 5 \text{ and } 5' = 1$$

If I tried to find out the complement of 2

If I compare 2 with 3

$$2 \wedge 3 = 1 \text{ but } 2 \vee 3 = 4$$

If I compare 2 with 4, $2 \wedge 4 = 2$ and $2 \vee 4 = 4$

" " 2 with 5, $2 \wedge 5 = 2$ and $2 \vee 5 = 5$.

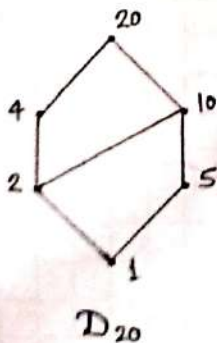
Similarly if I compare 3 with 2, 4, 5, 1, Similarly 4 with 5, 2, 3, 1 I will see that these 3 elements (2, 3, 4) do not have any complement.

- A Lattice, where every element has at-least one complement is called a Complemented Lattice.

So eg 1 & eg 2 have the examples of Complemented Lattice, but eg 3 is a non Complemented lattice.

Lets see more examples:

Ex.



It is a bounded lattice so I can directly write the $1' = 20$ and $20' = 1$.

If I try to check the $4'$, if I compare 4 with 20, $4 \wedge 20 = 4$ so not a complement.

If I compare 4 with 10, $4 \wedge 10 = 2$, so not a

complement, If I compare 4 with 5, $4 \wedge 5 = 1$ & $4 \vee 5 = 20$

$$\text{So } 4' = 5 \text{ \& \& } 5' = 4$$

If I try to compare 2 with 5, $2 \wedge 5 = 1$ but $2 \vee 5 = 10$,

If I compare 2 and 4, $2 \wedge 4 = 2$ and $2 \vee 4 = 4$,
not a complement.

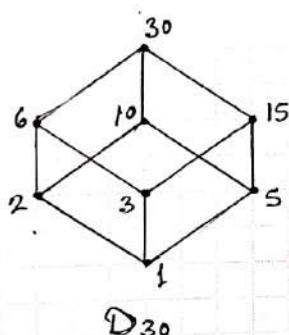
If I compare 2 and 20, $2 \wedge 20 = 2$ and $2 \vee 20 = 20$,
not a complement. So there is no complement of 2.

Similarly if you check, there is no complement of 10,
so two elements, 2 and 10 have not any complement.

$$2 \wedge 10 \neq 0 \quad (2 \wedge 10 = 2)$$

So D_{20} is not complemented lattice.

Ex.



$$1' = 30, \quad 30' = 1$$

$$2' = 15, \quad 15' = 2$$

$$5' = 6, \quad 6' = 5$$

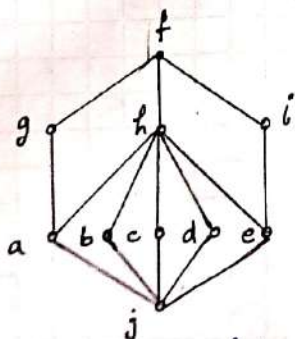
$$3' = 10, \quad 10' = 3$$

So each & every element is having at least one complement, so this is a complemented lattice.

Now, I leave it to you to actually verify that what I have written is right. How do you verify? If a & b are complement to each other then $a \vee b = 1$ and $a \wedge b = 0$

D_{30} is complemented lattice.

Ex.



Can you find the complement of 'a'?

Complement of a can't be b, c, d & e ,
as infimum is j but supremum is h .

In fact the complement of a is i .

Let's verify. $a \vee i = f$, $a \wedge i = j$

So $a' = i$ and $i' = a$

Now as a homework, I will leave it to you find complement of all elements and check if it is a complemented lattice?